

Lower Limb Amputation and Gait

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SUMMARY PEARLS

- Lower extremity limb loss is increasing over time and will double over the next 25 years.
- New surgical techniques, including osseointegration, and advances in prosthetics techniques have lasting impacts in prosthesis usage and functional independence.
- Prosthetic fit, alignment, and proper component choice are essential to the rehabilitation process.
- Pain management includes both the actual residual limb and the phantom leg.
- Early and persistent gait training with identification and correction of deviations using gait analysis provides lifelong advantages.
- Pediatric patients with limb loss will require more attention than adults.

Introduction

The diverse spectrum of patients that present for lower limb amputation rehabilitation services creates a practice environment that can be both exciting and challenging. This spectrum ranges from children with congenital limb deficiencies to elderly individuals with multiple medical conditions and amputations secondary to complications from vascular disease. The challenge with this broad spectrum of patients is twofold. First, the prosthetic restoration and rehabilitation approach must be individualized for each unique patient presentation. Second, the comorbidities associated with the amputation etiology vary and require a patient-centric approach to the management of these conditions. For example, children with congenital amputations may have associated genetic abnormalities. Young adults with traumatic amputations may have associated injuries, such as traumatic brain injury, and the elderly with vascular-related amputations commonly have significant medical comorbidities, including cardiovascular disease and chronic kidney disease. Advances in medical care, therapy approaches, and prosthetic technology have provided the ability for persons with lower limb amputations to achieve enhanced functional outcomes and quality of life. For these outcomes to be obtained, rehabilitation professionals practicing in this field must possess a wide spectrum of knowledge and skills ranging from dermatologic and wound care management to observational gait analysis. Advances in the sophistication of prosthetic technology and rehabilitation interventions also require that providers have detailed understanding of these advances to appropriately apply this specialized technology in the clinical setting. Because of the highly technical and specialized nature of this knowledge, patients are best served when this care is provided by a coordinated interdisciplinary team.

This chapter examines the comprehensive rehabilitation management of persons with a lower limb amputation. The goal of this chapter is to highlight the rehabilitation principles and strategies that can be applied in the clinical setting across the continuum of care. Topics covered in this chapter include epidemiology, amputation-level terminology, functional classification, and implications of surgical techniques. Management considerations in the areas of medical care and prosthetic limb restoration are reviewed with an emphasis on understanding how to ideally match the characteristics of a prosthetic device to the functional needs of each patient. The chapter also includes information on both normal gait and the evaluation and management of prosthetic gait deviations. The chapter concludes with a look into the future of this field.

Epidemiology

Despite initial declines in amputation rates at the turn of the millennium because of advances in medical interventions and an emphasis on prevention programs, studies have shown that the rate of amputation procedures and the prevalence of persons living with limb loss continues to increase. For example, the reported rate of nontraumatic amputations per 1000 people with diabetes reduced by roughly half between 2000 and 2009, from 8.5 to 4.4 persons; however, since 2009, these rates have increased slightly, to 4.8 persons. These trends were observed across most age, sex, and race/ethnic groups, with higher rates observed in the oldest age groups, Blacks, and males. These increases in nontraumatic amputation were driven by increases in the rates of toe and foot amputations, while major lower limb amputations at the transtibial and transfemoral levels continue to decline.⁴⁴ The current prevalence for those living with amputation in the

United States is reported at 2.3 million, with the majority of these being amputations of the toes, foot, and leg (lower and upper). The data do not include those individuals who are uninsured or who are covered by the US Department of Veterans Affairs (VA) or the US Department of Defense Military Health System (TRICARE).^{11,41} Between 2001 and December 2018, the military conflicts in Iraq and Afghanistan resulted in 1722 US service members sustaining one or more conflict-related extremity amputations.⁸¹ A much larger number of amputations are performed each year in VA medical facilities, with an average of 9114 new amputation procedures performed annually between 2013 and 2018.⁹⁰

The prevalence of individuals living with limb loss is anticipated to continue to increase in the future. This trend is related to a number of factors, including the aging of the overall population and an increase in the incidence of diabetes. Despite enhanced procedures for limb revascularization, the incidence of amputation in those with peripheral vascular disease does not appear to be declining.³⁴ Studies have predicted a doubling of the overall elderly dysvascular amputation population by 2030 and that the overall amputation prevalence will double between 2005 and 2050.^{31,101} The majority of persons with amputations have acquired their limb loss as a result of a disease process, such as diabetes or peripheral vascular disease. Amputations secondary to vascular conditions and diabetes have been reported to account for 82% of limb loss discharges, and 97% of vascular-related amputations involve the lower limb.²² Trauma-related amputations accounted for approximately 16% of amputations, and those resulting from malignancy and congenital deformity were responsible for approximately 1% of amputations each.²² Diabetes has been found to increase the risk of amputation to a greater degree than either smoking or hypertension. Diabetes is reported to contribute to 67% of all amputations.⁷³ The age-adjusted amputation rate for persons with diabetes has been found to be 18 to 28 times greater than that of persons without diabetes.⁸⁹ Among people with a lower extremity amputation, smoking cigarettes has been associated with a reamputation risk 25 times greater than that of nonsmokers.⁵⁴

The most frequent level of amputation in the lower extremity also varies according to the etiology of the amputation. As noted previously, the majority of lower extremity amputations are related to vascular diseases. Toe amputations are the most common level of amputation overall when counting both major and minor amputations.⁹⁰ Although associated with higher rates of reamputation, advances in limb salvage techniques have resulted in an increased number of partial foot amputation procedures being performed over the past 15 years.⁸⁵ Most sources agree that amputation at the transtibial level is the most common major amputation level in the lower limb with transfemoral amputations being the second most common major amputation level.

Amputations caused by disease processes generally occur in the aging individual and are associated with numerous comorbidities, such as cardiovascular disease, hypertension, end-stage renal disease (ESRD), and arthritis. The comorbid conditions associated with small but significant compromise to prosthetic mobility are individuals aged over 65 years and histories of stroke, anxiety and panic disorders, and peripheral vascular disease. By contrast, when controlling for these known risk factors, the presence of diabetes, obesity, chronic obstructive pulmonary disease (COPD), and arthritis were not associated with reduced prosthetic mobility.⁹⁹

Persons with amputations caused by trauma, including military-related injuries, are predominantly younger in age and typically

require a longer continuum of care after their amputations. These individuals may have more specialized prosthetic and rehabilitation needs secondary to the increased likelihood of returning to work or high-level sports and recreational activities. Individuals with military-related traumatic amputations also require management of associated injuries, such as traumatic brain injury, hearing loss, visual impairment, and posttraumatic stress disorder (PTSD).

Survival rates after amputation are quite variable depending on the etiology of the amputation. Meta-analysis has reported pooled mortality rates after nontraumatic amputation at 33.7%, 51.5%, 53%, 64.4%, and 80% at 1-, 2-, 3-, 5-, and 10-year follow-up, respectively. Among diabetic patients specifically, 1- and 5-year mortality after amputation was 27.3% and 63.2%, respectively.⁵⁹ Comorbidities such as COPD, ESRD, active coronary artery disease with heart failure and angina, and older age are associated with increased mortality in these patients.⁷⁹ Individuals with traumatic amputations have been noted to have more significant cardiovascular and metabolic issues that appear to be related to their traumatic amputation and not accounted for by obesity, sedentary lifestyle, or tobacco use. Persons with traumatic amputations also have been identified as having after-injury increased rates of hypertension, ischemic heart disease, and diabetes mellitus. Despite these findings, individuals requiring an amputation secondary to a traumatic injury tend to have relatively normal long-term survival rates.

Terminology

The International Organization for Standardization (ISO) terminology for the description of both acquired amputations and congenital limb deficiencies has been widely accepted by clinicians, researchers, and professional organizations.⁷⁸ The classification of congenital lower limb deficiencies is further detailed later in this chapter (see Pediatric Lower Limb Loss). [Table 11.1](#) provides a comparison of the ISO terminology and the more traditional, common terminology, and a description of each level of acquired lower limb amputation. Although the common terminology is still used, the ISO terminology is recommended to improve the accuracy and consistency of amputation level description. The major and minor classifications of amputation levels is also included in [Table 11.1](#). Major amputations have traditionally included amputations that occur at or proximal to the ankle disarticulation level. Although still frequently used, the major and minor amputation terminology can be misleading because amputation levels classified as minor (partial foot and digit amputations) can still have significant functional and quality of life implications for the individual with the amputation.

Implications of Amputation Level and Surgical Technique

Both the level of amputation and the specific techniques used during surgery can have profound impacts on long-term functional outcomes after lower limb amputation. This impact can be expressed in terms of successful mobility, enhanced prosthetic socket comfort, and a reduction of skin breakdown complications. Although the primary goal of amputation surgery is removal of the diseased, damaged, or dysfunctional portion of the limb, the surgery must also result in a residual limb that is optimized for load bearing through the prosthesis, prosthetic mobility, motion control, and proprioceptive feedback to achieve the most successful

TABLE 11.1 Lower Extremity Acquired Amputation Classification Terminology and Description

International Organization for Standardization Terminology	Common Terminology	Description	Major Amputation
Hemipelvectomy	Hemipelvectomy	Removal of the entire lower limb and partial removal of pelvis	Yes
Hip disarticulation	Hip disarticulation	Amputation of the entire lower limb, including proximal femur	Yes
Transfemoral	Above knee	Amputation through the shaft of the femur	Yes
Knee disarticulation	Through knee	Amputation through the knee joint with retention of the distal femur	Yes
Transtibial	Below knee	Amputation through the shaft of the tibia	Yes
Ankle disarticulation	Syme	Amputation through the ankle joint	Yes
Partial foot	Chopart, Lisfranc, transmetatarsal, ray	Amputation through the structures of the foot (transverse or longitudinal)	No
Digit(s)	Toe(s)	Removal of one or more toes	No

outcomes. Amputation surgery should not be considered to be a procedure of last resort or a failure of care. Instead, amputation surgery should be viewed as a reconstructive procedure that has the potential to improve a person's functional independence, mobility, and quality of life.

Although advances in surgical techniques have made limb salvage a more viable option in certain circumstances, delaying definitive amputation with attempts at limb salvage that have a low likelihood of success can create negative consequences, such as greater pain levels, the need for multiple surgeries, and prolonged immobility with associated deconditioning. Several principles should be considered in relation to the impact of amputation surgery level on rehabilitation outcomes. It is generally recommended to preserve as much limb length as possible at the time of amputation surgery. Although this principle holds true in many situations, there are instances in which preserving additional length has no functional benefit and may actually result in a less optimal outcome. For example, performing a transtibial amputation in the distal third of the tibia is typically not recommended because this may limit the use of high-profile prosthetic feet, create challenges with prosthetic alignment, and the lack of soft tissue coverage in the distal third of the tibia can lead to decreased comfort, an increased risk of skin breakdown when wearing a prosthesis, and difficulty in accommodating the appropriate limit flexion in the socket. Partial foot amputations can provide the potential for short-distance ambulation without a device, but these amputation levels are difficult to fit with an adequate prosthesis and have a high rate of equinovarus deformity secondary to muscular imbalance, which potentially can be prevented by salvaging or reimplanting the tibialis anterior tendon. Further, the historic assumptions that partial foot amputation should be preferred over transtibial amputation because it provides a more normalized gait with reduced energy expenditure and improved quality of life have been challenged in recent literature.²³

Disarticulation-level amputations may provide sparing of additional limb length and can provide the advantage of continued bone growth in those who are skeletally immature at the time of amputation. Amputations at the ankle disarticulation level afford the potential to ambulate short distances without a prosthesis.

Knee disarticulation-level amputations also provide the potential advantages of distal weight bearing, self-suspension, and a longer lever arm for greater prosthetic limb control. Disarticulation-level amputations may also be favored at times in individuals with spinal cord injury to maintain muscle balance and reduce the risk of contracture formation. However, despite these apparent advantages, disarticulation-level amputations are often not recommended because they can result in poorer cosmetic outcomes and because they may limit the availability of prosthetic component options.

Although the ultimate level of amputation is often dictated by the amount of blood flow and tissue viability in cases of vascular disease or by the extent of soft tissue and bone damage in cases of trauma, it is ideal for rehabilitation providers to add input regarding the amputation level before the time of surgery when circumstances allow. Prediction of healing requires careful evaluation of multiple variables, including nutritional status and tissue perfusion. As noted, in persons with vascular disease, it is desirable to preserve length; however, performing the amputation at a more proximal level, where the likelihood of timely and successful healing is greater, may be a better option to facilitate rehabilitation and avoid multiple surgical interventions. In cases where there is a need for amputation secondary to cancer, preserving length always has to take lower priority to preserving the person's life. In cases of extremity trauma, advanced surgical techniques with bone growth stimulation and tissue expanders have resulted in greater opportunities for both limb salvage and limb length sparing in cases of amputation. The long-term outcomes of limb salvage compared with amputation after extremity trauma remain mixed,⁹ although studies of US military personnel have shown improved overall functional outcomes (with the Short Musculoskeletal Function Assessment questionnaire) in those with amputations compared with those with limb salvage.²⁵

Despite a lack of research evidence supporting the utilization of one particular surgical technique over another, the surgical technique used at the time of amputation can also have a lasting impact on successful prosthetic limb use.⁹¹ Amputation surgery techniques should provide an adequate amount of soft tissue padding for a comfortable interface with the prosthetic socket while

avoiding excessive, redundant soft tissues that can make donning the prosthesis difficult and allow excessive motion between the residual limb and the prosthetic socket during ambulation. Preserving skeletal length without adequate soft tissue coverage can lead to recurrent skin breakdown, soft tissue infections, osteomyelitis, and the need for revision surgery. Surgical techniques should also strive to avoid the development of adherent scar tissue over distal bone, which can lead to both pain and recurrent skin breakdown. Surgical technique can also have implications on the development of painful neuromas and heterotopic ossification (HO) in the residual limb.⁸⁰

Surgical management of the remaining muscular structures is also important. Myofascial closure involves closure of the muscle fascial envelope without attachment to the bone. This may provide adequate cushioning over the distal bone, but it provides limited stabilization of the muscle structure and may result in limited muscle power generation. Myoplasty techniques involve suturing of the muscle fibers and fascia. This may enhance muscle stability but can also result in a mobile sling of muscle that creates excess movement and the potential formation of painful bursa. With myodesis techniques, the muscle and fascia are directly sutured to the periosteum of the bone. This provides greater stabilization of the muscles and can enhance the contractile effectiveness and efficiency of the muscle. Adductor myodesis procedures to achieve muscle balance are especially important after transfemoral amputation to avoid excessive femoral abduction both in standing and during ambulation.⁸⁰

An advanced amputation surgical technique, agonist-antagonist myoneural interface (AMI) or Ewing amputation procedure, has been trialed in individuals with transtibial-level amputations. The AMI procedure preserves the dynamic muscle relationships within the native anatomy, which facilitates proprioceptive signals from mechanoreceptors within both muscles to be communicated to the central nervous system. When these signals are communicated to a prosthetic ankle joint, subjects have demonstrated improved control over the prosthesis compared with subjects having traditional amputation.¹⁴

Additional advancements in amputation surgery have been made in the area of nerve management. The traditional approach of ligating nerves under tension so that any subsequent neuromas will be buried deep within the limb is increasingly being replaced with more thoughtful surgical techniques. Targeted muscle reinnervation (TMR) involves the thoughtful coaptation of severed sensory nerves into the nerves of the residual muscle bellies. Although originally developed to increase the number of surface myoelectric sites in upper limb amputations, the procedure has demonstrated the ability to reduce both nerve pain and phantom limb pain (PLP) in both upper and lower limb amputations.⁵² A related technique is the regenerative peripheral nerve interface (RPN), whereby the incised end of the nerve is implanted into a free muscle graft. This technique can help for both prevention of neuromas and systems of prosthetic control.

Residual Limb and Skin Care

Skin care is an area of paramount importance that requires consistent effort on the part of caretakers and the individual with an amputation. The latter should form strong, early habits for at least daily inspection of skin of the residual limb and the resolution of skin problems as soon as they develop. No condition should be considered too trivial to treat. If neglected, a skin disorder has the potential to progress and cause much greater problems, such as

sepsis and further surgical revisions. The use of a prosthesis subjects the residual limb skin to numerous physical stressors. Suction sockets and other suspension systems can create both positive or negative pressures, and mechanical stresses such as shear forces, friction, excessive loading pressures, humidity, sweating, and stretching can cause skin problems.

General residual limb care recommendations include cleaning the residual limb daily, preferably in the evening, with soap and water and then patting dry. Washing at night prevents the hydrated skin from excessive shear and friction stress with prosthesis use. When the patient is not wearing the prosthesis, a shrinker or an ACE wrap should be applied to minimize or decrease swelling. If a gel liner is used, nighttime washing of the liner with soap and water is recommended. This allows for drying of the liner overnight to avoid any moisture when the liner is placed over the residual limb, which can provide an environment for bacterial flora. After prosthetic limb wear, the residual limb should be examined for irritation, breakdown, blistering, or erythema. If any of these exist or erythema does not resolve within 20 minutes, the prosthesis should not be worn, and a clinical professional should be consulted within the next few days. As the prosthesis is worn throughout the day, socks should be added to ensure an appropriate fit is maintained. The amputee should make sure that the anatomic points of the residual limb line up appropriately with relevant points on the prosthesis (e.g., fibular head with the recess for this in the socket and patellar tendon with the patellar tendon bar). The skin should be examined once or twice a day, including the use of a mirror if there are areas, such as the distal end, that are not easily viewed.^{53,57,63}

This type of close monitoring and care should also be applied to the contralateral limb. Daily cleansing, drying, and close inspection should occur, particularly for areas difficult to assess (such as between the toes, plantar surfaces of the foot, and heel). Frequent assessment should also include sensation, pulses, edema, temperature, and examination for any evidence of trophic or motor changes. Contralateral amputations are common, with one 2018 population study in Ontario, Canada, finding one in six patients who survive a lower limb amputation ultimately ending up with a contralateral amputation over a period of 10 years, regardless of etiology.⁴⁵ Thus aggressive preventive measures are warranted. Podiatric care of corns, calluses, and nails is also helpful in the prevention of complications. A baseline visual and sensory examination is recommended during routine checkups to assess the patient's capability in self-care; if needed, caregivers should be educated on routine skin care.

Despite the skin care recommendations, dermatologic issues are common in the amputee population. Several studies have revealed dermatologic conditions in 34% to 70% of lower extremity amputees.^{16,27,56} Risk factors for developing dermatologic conditions in lower extremity amputations included level of amputation, employment status, absence of peripheral vascular disease, and use of a walking aid. Overall, the highest risk factor is a transtibial level of amputation. The study authors ascribed these risk factors to an overall increase in prosthesis use and time spent ambulating.²⁷ In another study of lower extremity prosthetic users, five skin conditions comprised 79.5% of the observed skin problems. These conditions included irritations, ulcers, inclusion cysts, verrucous hyperplasia, and calluses.²⁶ In addition to these conditions, other reports indicate the following frequently occurring conditions: allergic contact dermatitis, acroangiodermatitis, epidermal hyperplasia, follicular hyperkeratosis, bullous disease, infections, and malignancies.⁶² Early in

the use of a suction socket prosthesis, the amputee may experience capillary hemorrhage, edema, and reactive hyperemia. This is generally benign and self-resolving. If edema persists, shrinkers, ACE wraps, or even an intermittent oral diuretic may be used to allow the edema to resolve.⁵⁴

One study found that ulcers were the primary dermatologic complaint in transtibial amputees.²⁷ Poor socket fitting, along with improper liner and sock use have been associated with ulcer formation. Amputees with vascular insufficiency (e.g., from diabetes mellitus) had a higher risk of chronic ulcers. Treatment includes removal of the irritant via socket modification and education on sock and liner management. Wound care to optimize healing and reduce bioburden is important but outside the scope of discussion for this section. Decreased or complete cessation of prosthesis use may be necessary during wound healing. Occasionally, oral or even intravenous antibiotics may be required for treatment. Prolonged healing may necessitate more invasive interventions, including surgical debridement or revision.⁴

Allergic contact dermatitis, an erythematous, weeping, and pruritic rash, representing up to one-third of the dermatoses seen in prosthetic wearers, is often caused by the prosthetic materials.⁶² Patch testing should occur with the first round of testing, including standard allergens, components of the prosthesis, topical medications being used, and locally applied cosmetics and moisturizers. If the first panel is negative, then further testing can extend to adhesives and additional cosmetics. Because of the difference in construction among different prosthetic providers, identification of the allergen should be done in association with the local prosthetist. For example, nickel can be present in metal fastenings. Leather used for attachments may have chrome elements. Rubber or synthetic rubber materials may be present in suction socket valves. Treatment consists of elimination of the specific allergen and use of topical and/or oral steroids.^{16,57}

If residual limb skin care or liner care is insufficient, the residual limb and liner provide an ideal environment for fungal and bacterial overgrowth. One study of residual limb skin identified *Staphylococcus epidermidis*, *Staphylococcus aureus*, and *Streptococcus* spp. as the most common bacterial flora.⁴⁶ *S. aureus*, *Trichophyton rubrum*, and candidal infections were present as folliculitis, furuncles, or superficial skin infections. Management includes more stringent skin care, topical antiseptics or antibiotics, and occasional systemic antibiotic/antifungals.⁵⁷

When there is excessive socket pressure proximally and inadequate distal contact, verrucous hyperplasia can develop.⁵ This condition has a characteristic appearance consistent with its name “verrucous” or warty. Vascular injury or chronic bacterial infection can also play a role in its development. Shrinker socks and modification of the socket to apply appropriate pressures to the distal end help to resolve this problem.

Inadequate socket fitting and alignment can also result in bursitis. Synovial bursae are fluid-filled sacs that facilitate the movement between muscle and bone, ligaments, and/or tendons and develop during intrauterine life. Adventitious bursae develop from excessive shearing forces of the skin, particularly over bony surfaces, affecting 7.2% of residual limbs.³² These shearing forces cause a breakdown of fibrous connective tissue with mucoid and myxomatous degeneration. There is no true synovial endothelial lining.³² Bursitis results from either acute or chronic inflammation. On examination, the bursa is a fluctuant, painful swelling, generally over areas such as the fibular head, tibial tubercle, patella, or end of the residual limb and can be associated with overlying erythema.¹⁶ If needed, diagnosis can be confirmed through

ultrasound or magnetic resonance imaging. The first line of treatment is usually a modification of the prosthesis. If there is suspicion of an infection, the bursa should be aspirated and the aspirant sent for analysis and culture.

Epidermoid inclusion cysts occur when elements from the epidermis are implanted in the dermis from repeated pressure and friction from the prosthesis. The cells within the cyst produce keratin, and the cyst can drain intermittently. These can be asymptomatic. If they do become symptomatic, they can present as small masses (follicular keratoses), which may also become infected and painful. Treatment consists of excision, incision and drainage, and the use of antibiotics. When asymptomatic and unproblematic, these can be left alone. To prevent the condition from worsening, liner material can be adjusted to provide smooth motion of the socket wall on the skin, and topical glucocorticoids can be used at night to reduce inflammation.^{4,18,57}

Another common problem is hyperhidrosis, or excessive sweating, of the residual limb with use of the prosthesis. This situation can hinder the use of the prosthesis, from thermal discomfort, difficulty in socket fitting, or as a risk factor for development of dermatologic conditions.³⁶ Approximately 30% to 60% of amputees are affected,⁴⁰ and 53% of the amputee population in a literature review listed thermal and perspiration discomfort regardless of the prosthesis type or level of amputation.³⁶ Treatment includes altering liner material, socket design, topical antiperspirants, botulinum toxin injections, or electrical stimulation.^{10,40}

It is also important to consider that the residual limb is an extension of the rest of the body, and systemic dermatologic conditions, such as eczema, can involve the residual limb as well. Skin conditions that exhibit the Koebner phenomenon, with the appearance of lesions on sites of trauma such as lichen planus or psoriasis, may be present on the residual limb with minimal involvement of the rest of the body, given the unique skin-liner environment.⁵⁷

Pain Management

Postamputation pain can range widely in both severity and persistence. The two fundamental types of pain are residual limb pain (RLP) and PLP. RLP involves pain that is restricted to the anatomic region of the residual limb. PLP involves pain that is perceived in the portion of the limb that is no longer present. This latter type of pain has an estimated prevalence of up to 80%. It occurs mostly after surgery, but its onset can occur at later times. Episodes can last from seconds to minutes to hours. Generally, PLP tends to diminish within 6 months after surgery.²⁸ In about 30% to 40%, the pain is severe; it is moderate in 25%. Only around 10% continue to have severe pain 6 months or more after surgery. RLP occurs more frequently in patients with PLP than in those who do not have PLP. Pain from the residual limb can also appear to radiate into the part of the limb that is no longer present.

Somatic pain in the residual limb can originate from a variety of sources, including HO, infection, tumor, ischemia, neuromas, or arthritic joint changes. Infection may be superficial or occur in deeper tissues and lead to the development of osteomyelitis. Poor surgical technique that leaves bone improperly trimmed or muscle and fascia inadequately sutured can result in mechanical RLP that can be exacerbated by wearing the prosthesis. Treatment can include socket or prosthetic modifications or surgical revision of the residual limb. Bony overgrowth can occur in children and more rarely in adults. Growth in the distal end can result in an irregular area of bone that projects into soft tissues, with the potential for

causing pain and skin breakdown with prosthetic use. Socket modifications are attempted as a first-line effort to manage this situation, and surgical revision can be done if these efforts fail.

If there are no painful sensations but there are still feelings and sensations in the portion of the limb that is no longer present, then this is called phantom limb sensation (PLS). PLP can be perceived in any part of the missing amputated limb but more often occurs in the more distal parts such as the foot. The quality of pain can be variable and described as dull, squeezing, cramping, electrical like, shooting, or sharp. It commonly manifests in an episodic manner with a severity that ranges from mild to severe and incapacitating. PLP tends to occur within the first few months after amputation and can persist indefinitely. The reported prevalence ranges are up to 80% in the first years after surgery.¹³ Supraspinal, spinal, and peripheral mechanisms are thought to play a role in the origin of phantom sensations and PLP. Some findings²⁸ point to a reorganization of the somatosensory cortex around the area representing the amputated part. Treatments can be directed at modulating the activities at any of these levels.

PLS is nearly universal in the early recovery period postamputation. RLP can further be classified into either neuropathic or somatic origins. Neuropathic origins include neuromas and complex regional pain syndrome. On dissection, neuromas demonstrate growths of Schwann cells amidst proliferating axons all encased within scar tissue. The free ends of the axons exist without Schwann cells, and the axonic environment of the scar tissue can create conditions in which the free nerve endings may fire repetitively.¹³ Virtually all amputees have neuromas at the site of the amputation, yet only 10% to 15% have pain from these neuromas. Diagnosis is confirmed by the presence of appropriate signs and symptoms. The pain from a neuroma is generally aching, cramping, or shooting with an intermittent, episodic nature. Provocation with pressure at the site of the neuroma helps confirm the source. Treatment options consist of physical modalities, such as acupuncture, socket modifications, ultrasound, massage, vibration, and percussion.⁹⁶ Ideally, the management of PLP should include an interdisciplinary team to implement a multimodal system of treatments.²⁸

Many treatment options have been tried to control PLP (Box 11.1), even though there are few controlled studies to provide guidance in this area. Efforts can be directed toward prevention of PLP and treatment through pharmacologic, nonpharmacologic, and surgical measures. Control of preoperative and immediate postoperative pain can help in the prevention and reduction of PLP.²⁸ Categories of pharmaceutical interventions include *N*-methyl-*D*-aspartate (NMDA) receptor antagonists, opioids, anticonvulsants, antidepressants, local anesthetics, and calcitonin. The NMDA receptor antagonists, such as ketamine, memantine, and dextromethorphan, are thought to exert their effects at the dorsal horn. One review of the pharmacologic management of PLP showed that only the NMDA receptor antagonists consistently provided pain relief. Within this class, ketamine demonstrated control of pressure pain thresholds and the windup phenomenon (IV bolus of 0.1 mg/kg/5 min and subsequent infusion of 7 µg/kg/min).³⁹ Opioids operate at both the spinal and supraspinal levels. Calcitonin exerts its effects centrally. The effectiveness of both calcitonin and anticonvulsants varies among studies.²⁸

Psychological treatments, such as guided imagery, biofeedback, and hypnosis, have been aimed at altering negative emotions, increasing adaptation to pain, and adjusting body image. Mirror

• BOX 11.1 Treatments for Phantom Limb Pain

- Pharmacologic
 - Opioids
 - Oxycodone
 - Hydromorphone
 - Morphine
 - Antidepressants
 - Imipramine
 - Mirtazapine
 - Amitriptyline
 - Nortriptyline
 - Anticonvulsants
 - Gabapentin
 - Carbamazepine
 - *N*-methyl-*D*-aspartate receptor antagonists
 - Dextromethorphan
 - Memantine
 - Ketamine
 - Miscellaneous
 - Clonidine
 - Mexiletine
 - Calcitonin
 - Tramadol
- Injections
 - Lidocaine
 - Corticosteroid
 - Botulinum toxin
- Complementary
 - Transcutaneous electrical nerve stimulation
 - Mirror therapy
 - Acupuncture
 - Hypnosis
 - Cognitive behavioral therapy
 - Virtual reality
- Surgical
 - Neuromodulation
 - Peripheral nerve reconstruction

therapy has some of the strongest research-based support. In this treatment, a mirror is placed adjacent to the intact limb, and then the intact limb is moved in exercises designed to promote reorganization of the cortex with this visual input.²⁸ Virtual reality systems have been used as an alternative to using mirrors.

Multiple electrical stimulation techniques have also been studied. Transcutaneous electrical nerve stimulation has shown promise. There are several studies that indicate effectiveness with placement of the electrodes on the intact limb.⁸⁸ More invasive modalities have involved dorsal root ganglion stimulation, spinal cord stimulation, motor cortex stimulation, and deep brain stimulation. Other centrally oriented treatments have included transcranial direct current stimulation and repetitive transcranial magnetic stimulation.¹⁸ Pulse radiofrequency energy treatment was reported as successful in one case study.³²

Surgical excision can also be performed but runs the risk of creating a new, painful neuroma. TMR and RPN, discussed earlier in this chapter, have emerged as new tools to control the pain from neuromas.

In addition to the development of neuromas, entrapment of nerves within scar tissue at the surgical incision site can occur with resulting pain. Shear, pressure, and traction forces from the prosthesis can either evoke or worsen this pain. The socket can be modified to decrease pressures in this area or redistribute it. If this

is ineffective, then injections into the scar or oral medications can be used. Surgical excision of this area is generally not effective.

Psychological Support

An amputation and the associated changes in body image and functional capabilities can have a strong emotional impact that requires a period of adjustment and supportive interventions. Naturally, this adjustment involves not just the amputee but also the person's relationships and roles with regard to family and friends. The amputation of a body part has been compared with the loss of a loved one, and some have described the psychological process in three phases.⁶⁷ In the first phase, combined feelings of shock, confusion, and numbness lead to a general feeling of emptiness. Daily tasks can be overwhelming. In the second phase, mourning predominates and consumes most of the amputee's energy and focus. Finally, the amputee progresses to the adjustment phase and finds a sense of self-worth and competency in daily life. Several factors may hamper progression through the phases to successful adjustment. Such factors include insufficient support from family members and caregivers; negative emotional states, such as a feeling of social isolation, low self-esteem, and a lack of a sense of wholeness; social anxiety; and body image discomfort.⁴⁸ The level of anxiety and depression is higher in individuals with traumatic amputation. Rehabilitation appears to improve employment rates but not necessarily the rate of substance abuse or the quality of relationships.⁵⁸ The level of anxiety and depression tends to decline after 2 years. In amputees, social discomfort and body image anxiety have been associated with activity restriction, depression, and anxiety. In contrast, the following characteristics have been associated with improved limb-loss adjustment: increased time since amputation, social support, prosthetic satisfaction, active coping attempts, lower amputation levels, and lower degrees of RLP and PLP.⁴³ In a study examining the relationship among tenacious goal pursuit (TGP), flexible goal adjustment (FGA), and affective well-being in individuals with lower limb amputation, TGP and FGA had different relationships with subjective well-being. TGP signifies changing one's life situation or behavior to facilitate goals, whereas FGA signifies changing goals to accommodate situational limitations.¹⁵ TGP seemed to foster a positive effect, whereas FGA played a role in reducing negative affect. According to the authors, these two factors could potentially serve two useful functions: identifying amputees who might have negative affective outcomes and providing useful areas of intervention to ameliorate negative affect.

One study evaluated the roles of positive attitudes for amputations, among other diagnoses, and found that hopefulness correlated positively with functional outcomes and participation during the inpatient rehabilitation program. Another study of young traumatic amputees reported that more than half had been given a formal psychological diagnosis. The most frequent included PTSD, anxiety, depression, and substance abuse, with some individuals having two or more. These psychological disorders have the capacity to impair adjustment to physical issues. Almost two-thirds of individuals with combat-related amputations have PTSD; anxiety and depression occur in approximately one-quarter; and substance abuse is found in approximately 6%. Level of amputation is associated with severity of psychological disorders.

In a study relating spirituality to quality of life of predominantly male subjects with traumatic transtibial amputations, the researchers concluded that "existential spirituality," female sex, and age greater than 50 years had a positive association with

increased quality of life.⁶⁹ Existential spirituality also correlated positively with satisfaction with life, health, and social integration. An important aspect of overall recovery for the person with limb loss is connecting with peer support and support groups. The largest group in the nation is the Amputee Coalition (<https://www.amputee-coalition.org/>), which provides support and information for all levels and time frames for people with limb loss and provides training to medical professionals and peer counselors. Psychological factors⁶⁶ such as depression, anxiety, and body image can impact quality of life and adjustment after lower limb amputation. Providing psychological support can help mitigate and alleviate these symptoms and facilitate the physical process of adapting to an amputation.

Studies examining return to work after lower limb amputation reveal that, overall, the rate of return to work is around 66%.⁸ Although the percentage of amputees keeping their preamputation job ranged from 22% to 67%, the employment achieved after amputation necessitated more education, had greater complexity, and required less physical functional requirements. General parameters that influenced return to work included age, sex, and educational level. Medical factors included level of amputation, number of amputations, comorbidities, the cause for amputation, and continued medical issues with the residual limb. Functional and prosthetic factors that related to return to work were time to fitting of the prosthesis, comfort with wearing the prosthesis, ability to walk distances, and other physical limitations with walking. Vocational factors influencing return to work include support from an employer, salary, who initiated the effort (individual, employer, family, agency), and a social support network.

Preprosthetic Phase Rehabilitation Considerations

The preprosthetic phase of rehabilitation begins after the amputation surgery and continues until the first fitting and training with a new prosthesis. Most patients fall within a certain time frame for fitting of the initial prosthesis and then the subsequent definitive prosthesis.⁵⁵ Before the surgery, the rehabilitative team, composed of the physiatrist, physical therapist (and potentially an occupational therapist), and prosthetist, should provide guidance and information about the rehabilitation process that will unfold after the amputation has been performed. Topics of interest to patients during this phase of recovery include wound management, pain experiences, PLS, expectations for rehabilitation, information on prosthetics, how to manage at home after amputation, financial information, availability of support groups in the community, and overviews of the recovery process.⁶⁸

Instruction on proper bed and chair positioning should be provided and an emphasis placed on appropriate exercises to maintain joint range of motion (ROM) as well as the strength and endurance of important muscle groups. Not all amputees will become prosthetic candidates, and this issue must also be approached carefully. Psychological support for the pending "loss" should also be provided.

When possible, providing input from a rehabilitation perspective on the length of preserved residual limb is important. Starting points on the transfemoral limb can be either the ischial tuberosity or greater trochanter, whereas for the transtibial limb the tibial plateau or tibial tubercle can be used. The endpoint is either the end of the bone or the distal soft tissue. For transtibial amputations, the optimal length of the residual tibia measured from

the tibial plateau is 3 to 6 inches.⁷² Too short a residual limb (i.e., <3 inches) will compromise control of the prosthesis, and too long (i.e., 6 inches) of a residual limb may limit the ability to use the posterior compartment musculature for soft tissue coverage over the distal residual limb. For the transfemoral amputation level, preservation of length must be balanced against displacement of the prosthetic knee center of rotation too far distal compared with the nonamputated limb.

After surgery, the primary emphasis is on wound care and healing, pain management, edema control, maintaining ROM in joints, initiating strength and mobility exercises, overall residual limb and prosthetic use education, and psychological counseling. Edema control of the residual limb can be achieved in multiple ways, including soft dressings, ACE wraps, semirigid dressings, rigid dressings, rigid removable dressings, plaster casts, and immediate postoperative fitting of a prosthesis, all of which not only control edema but also help reduce pain and protect the residual limb and surgical area from trauma. The choice of edema control depends on many factors, including the preference of the surgeon and the familiarity of the staff with the different options. Studies examining efficacy have revealed varying outcomes, but most authors^{46,72} have concluded that semirigid and semirigid removable dressings have greater effectiveness for edema control than elastic wraps/garments but only in the first few weeks. Additionally, evidence has suggested that the use of such semirigid removable dressing can protect the residual limb, including a reduced risk of conversions from transtibial to transfemoral amputation caused by fall-related limb damage during acute recovery.³⁰ In addition to edema control, shaping of the residual limb in preparation for prosthesis fitting is also important. Ideally, the transfemoral residual limb would evolve into a conical shape, whereas the transtibial one should be more of a cylindric one. Periodic circumferential measurements of the residual limb should be taken to assess volume and to help determine readiness for fitting. The transtibial residual limb has achieved a more mature shape when the distal end is slightly less in circumference than the proximal area. There is a more marked difference for the transfemoral residual limb.

Educating the preoperative and postoperative patient on appropriate position in bed and wheelchair is critical. In addition to assuring appropriate positioning in the bed and wheelchair to avoid flexion contractures of the hip and knee, monitoring of ROM in these areas should be performed regularly. When a prosthesis is not worn, transtibial amputees should keep the knee in full extension. Similarly, individuals with transfemoral amputations should be counseled not to put a pillow under the residual limb or between the legs when in bed to prevent the formation of joint contractures. Careful joint ROM measurements with a goniometer are important; with knee extension measured with the goniometer, the arms are carefully aligned with the femur and tibia, and hip assessment is performed with the Thomas test. Contractures of the hip and knee on the amputated side can hinder the process of fitting a prosthesis or prevent it altogether. A knee flexion contracture can increase the energy, strength, and endurance needed for prosthetic ambulation. A hip flexion contracture greater than 15 degrees makes prosthetic fit difficult, and appropriate alignment modifications to the prosthesis are required.

Preprosthetic mobility after surgery often involves the use of a wheelchair. With the loss of a limb, the center of mass (COM) has changed, and balance within a wheelchair can become more precarious if adjustments are not made. Because the COM moves posteriorly, the wheelchair can be made more stable by

also moving the axles of the posterior wheels posteriorly. This issue becomes even more significant for individuals with bilateral amputations. Antitip bars and weighting of the anterior seat posts can help in accommodating changes in the COM.

For those with very good single limb balance, ambulation with crutches may be a possibility; otherwise, the use of a walker can be useful for negotiating short distances. Energy consumption with walking aids requires more energy than ambulation with a prosthesis. Practice in the parallel bars may be necessary before sufficient competence has been developed. These skills in ambulation without a prosthesis are not a necessary prerequisite for prosthetic training and ambulation.

Prosthetic Training Phase Considerations

Aside from strengthening, ROM, and endurance exercises, early training for the amputee with a prosthesis should be based on an individualized therapeutic gait training program that may include tactile, verbal, and visual feedback during pre-, part-to-whole, and resisted gait interventions. These may occur during both traditional overground gait reeducation and where available treadmill training with or without body weight support is available and may be supplemented by psychological awareness training and functional gait activities, such as environment object negotiation.¹⁹

An additional component to early training in using a prosthesis concerns appropriate sock ply management. The patient must consistently apply these principles for as long as the prosthesis is used. The first principle for the amputee to understand is that the residual limb will vary in volume, and this requires an adjustment of prosthetic fit through the addition or deletion of socks. Prosthetic socks commonly come in 1-, 3-, and 5-ply types. With wear, the socks tend to lose some of their thickness. Factors that affect limb volume include fluid shifts associated with renal function or dialysis, muscle atrophy, weight gain or loss, and associated medical conditions, such as congestive heart failure. Wearing the prosthesis can create a pumping action that forces fluid out of the residual limb and reduces its size. During the first 3 to 12 months after amputation, the residuum will typically swell if a shrinker is not worn consistently. Socket changes are generally required when the amputee needs 15-ply (or more) socks to accommodate shrinkage of the residual limb. If the residual limb volume is stable for 8 to 12 weeks, then the time is appropriate for fitting of the definitive prosthesis, which usually occurs around 6 to 18 months after surgery.

Close observation of the patient during donning and doffing of the prosthesis can provide valuable information in determining issues with the fit of the socket. For the prosthesis with a pin lock system of suspension, the number of clicks gives some clue as to the level of fit. If the patient is only able to obtain a few clicks into the locking mechanism, then the patient may not be fitting down into the socket completely. If the speed of the clicks increases, this could indicate that the residual limb has shrunk and that additional socks should be added. Ease of donning is also a clue. If the prosthesis can be slid onto the residual limb easily and without resistance, then more socks are needed. Additionally, red marks on the residual limb can also give some indication of the fit. If the distal end of the limb is getting red and sore from too much pressure, then more socks are needed. If, after walking with the prosthesis, there are reddened areas (reactive hyperemia) that do not resolve after a few minutes, then the socket is creating excessive pressure on the residual limb in these locations, and adjustments have to be made. Alternately, in suction socket environments,

localized distal redness may indicate negative pressure caused by a lack of adequate distal contact. This distinction between excessive positive vs. negative pressure can be challenging and should be approached thoughtfully.

After the patient understands the importance of sock ply management and how to appropriately adjust this aspect, instruction in donning and doffing the prosthesis can occur. If a liner is being used, the liner must be applied with careful attention to technique. The liner should be turned inside out, and the distal end of the residuum should be placed directly against the liner, which should then be rolled upward without any air pockets. An air pocket between the distal end of the residuum and the liner can create suction and affect the skin accordingly. Liner care should include washing it with soap and water at the end of the day. Drying should occur with the fabric side directed outward. This position will protect the tacky inner area from trapping dirt, dust, and hair that can adversely affect the skin. Spraying the inner part of the liner with diluted rubbing alcohol once or twice a week can reduce the buildup of bacteria on the liner surface and help prevent residual limb infections.⁶³

Socks must be donned in a manner that eliminates wrinkles because these will create pressure areas that can cause skin breakdown. Wrinkles can be smoothed away with the tips of fingers. For the wearer of a transtibial prosthesis, the socket must be correctly aligned so that the bony prominences fit into the reliefs that were designed to accommodate them in the socket. The use of too many or too few socks can also prevent the correct seating of the residuum in the socket. In general, the socket should be donned in a consistent manner irrespective of the type of socket.

Once the correct donning technique has been mastered, the amputee can embark on other more advanced pregait activities. These exercises focus on fostering good balance and strength, weight shifting, and isolated parts of the gait cycle. Early training involves static weight bearing, dynamic weight-shifting exercises, reaching activities, repeated stepping actions in all directions, as well as identification and elimination of gait deviations. Tap-ups represent one activity that generates skill and confidence in weight shifting and weight bearing on the prosthesis. Within the parallel bars, the first steps in learning the various parts of the

gait cycle can begin. One of the first exercises involves learning the heel-toe pattern with the prosthetic leg. For the transfemoral amputee, this exercise will also initiate the learning process for manipulating the prosthetic knee. Ultimately and according to the patient's capabilities and certainly staying within any safety considerations, the patient may advance to walking on level surfaces in different settings and on uneven terrain. Negotiating environmental obstacles, such as ramps and curbs, will also be important.

Functional Classifications

The Centers for Medicare and Medicaid Services published a functional classification system for individuals with amputations to guide prosthetic limb prescription based on the actual or potential functional abilities of the person.¹² The guideline divides functional mobility into five categories and provides recommendations for the prescription of prosthetic components based on the functional mobility category (Table 11.2). These five categories have been referred to as the Medicare Functional Classification Levels (MFCLs), the K-Level Modifiers, or the Functional Index Levels. Although the amputee's functional mobility category determination should, to the extent possible, be based on objective clinical findings, the classification also allows for clinical judgment by the prescribing physician or team in predicting the anticipated functional level of the new amputee once the person has been fit with the prosthesis. The determination should also consider the individual's medical conditions and medical comorbidities that could affect the person's ability to function with the use of a prosthetic limb. Classification may be enhanced by having patients complete functional performance outcome measures, such as the Amputee Mobility Predictor.²⁴ The patient's goals and desires for prosthetic use must be considered as part of the prescription process, and if the goals of the patient are not realistic with respect to the benefits of a prosthetic limb, education in this regard will be required. It should be emphasized that the final determination of the prosthesis prescription is ideally a team decision involving the physician, prosthetist, therapist, and patient.

TABLE 11.2 Medicare Functional Classification Level Descriptions and Prosthetic Component Recommendations for Each Level

Functional Index Level	Description	Recommended Prosthetic Components
K0	No ability or potential to ambulate or transfer with use of a prosthesis, and prosthesis does not enhance the quality of life	None for function Potential for cosmetic prosthesis
K1	Ability or potential to transfer or ambulate with a prosthesis for household distances on level surfaces at a fixed cadence	Feet: solid ankle cushion heel, single axis Knees: manual locking, weight-activated stance control
K2	Ability or potential to ambulate limited community distances and traverse low-level environmental barriers; ambulation at a fixed cadence	Feet: multiaxial and flexible keel feet Knees: weight-activated stance control
K3	Ability or potential to ambulate unlimited community distances and traverse most environmental barriers; ambulation with variable cadence	Feet: multiaxial, energy storing Knees: hydraulic, pneumatic, and microprocessor controlled
K4	Ability or potential to exceed normal ambulation activities and use a prosthesis for activities exhibiting high impact, stress, or energy levels	Feet: energy storing or other specialty feet Knees: no specific limitations

• BOX 11.2 Prosthetic Prescription Essential Elements

- Socket
- Interface
- Suspension
- Pylon/frame
- Foot and ankle
- Knee unit if knee disarticulation or above
- Hip joint if hip disarticulation or above
- Extras (rotators, covers, etc.)

Prosthetic Restoration

Essential Elements of the Prosthetic Prescription

The prescription of a lower extremity prosthesis should be approached in an organized fashion to ensure that the essential elements are included in the prescription. Box 11.2 provides a template that can be used for any level of lower extremity prosthetic prescription. Depending on the level of the amputation and the functional goals of the individual with the amputation, not all elements of the template will be required. Determination of the patient's current and anticipated functional abilities and subsequent classification within the MFCL, as described, is important in the development of the prosthetic prescription. With the prosthesis prescription, the emphasis should be on identifying the best-suited class of prosthetic components to achieve the patient's functional goals rather than emphasizing the specific product name for each component of the prescription.

During the prosthesis prescription process, it should be recognized that many factors influence which specific components should be selected for each individual patient. Residual limb length, muscular strength, balance, coordination, vision, and motor control all impact stability during prosthetic ambulation and may require added stability to be incorporated into the prosthesis. The quality of the residual limb skin should be considered in selecting the most appropriate prosthetic suspension and interface system. Hand function, vision, and cognitive abilities need to be considered in regard to donning and doffing the prosthesis. In addition, the amputee's weight can limit the available component options. Lastly, variables such as prosthesis durability, reliability, cosmesis, and cost need to be part of the evaluation to determine the ideal prosthesis prescription. Ultimately, determination of the prescription should be a team decision involving the physician, the prosthetist, the therapist, and most importantly, the patient. The goal is to educate patients and their families about reasonable, available options and their advantages and disadvantages so that patients can contribute to an informed decision.

Socket Designs

The prosthetic socket serves as the platform for connecting the amputated residual limb to the prosthetic limb. In some circumstances, there is direct contact between the residual limb skin and the prosthetic socket, whereas in other circumstances, a liner or other materials are used as the interface between the residual limb and the socket. The socket itself can also function to suspend the prosthesis to the residual limb, but in most instances the suspension is accomplished through an additional suspension method. To achieve an ideal functional outcome with use of a prosthetic limb, the socket must be comfortable and secure as well as facilitate motion transfer

from the residual limb to the prosthetic limb. Having a secure and comfortable connection allows the amputee to effectively bear weight through the prosthesis and efficiently advance the limb during the swing phase of gait. Prosthetic sockets are made from various materials, and there are a variety of socket designs that have been developed for each level of lower limb amputation. The most commonly used prosthetic socket designs are highlighted next.

Transtibial Socket Design

There are several different transtibial socket designs that are currently being used in the prosthetics field. Some of these designs are considered hybrid designs because they combine the features of one or more traditional designs. Independent of the type of design, the goal is to provide a socket that is well fitting, comfortable, and secure.

Patellar Tendon Bearing

The major weight-bearing area for the residual limb in this design is at the patellar tendon with a counterforce in the popliteal region. Current socket technology also provides total surface contact with specific weight-bearing areas in the soft tissue regions of the residual limb. These would include the anterior muscle compartment, medial tibial flare, shaft of the fibula, gastrocnemius muscle, and distal end, which receives slight pressure. Excessive contact over bony areas is avoided. For the patellar tendon-bearing (PTB) socket, the anterior trim line is proximal to the patellar tendon insertion on the patella. The medial/lateral trim lines are at the midlevel of the medial condyle. The posterior trim line is at or just below the midpatellar tendon with reliefs for the medial and lateral hamstring tendons (generally at the bend of the knee so that the patient can bring the prosthesis to a 90-degree angle). Traditional bench alignment includes approximately 5 degrees of flexion in the socket.

Total Surface Bearing

This socket design creates more equal weight-bearing distribution throughout the socket. Unlike the PTB design, the total surface-bearing (TSB) socket is designed to globally apply forces throughout the residual limb. There is very little pressure difference between areas and limited relief for bony areas. Many TSB designs use gel-type interfaces, which also facilitate the more even distribution of forces over a larger surface area. Trim lines are similar to the PTB design. Clinically, the typical transtibial socket design is a combination of both the traditional PTB and the TSB design.

Transfemoral Socket Design

As with transtibial socket designs, there are many versions of the transfemoral socket. Newer socket designs have flexible inner liners with rigid external frames that can be fenestrated to allow for bony relief and muscle contraction. The external frames that attach the socket to the distal prosthetic components are laminated with carbon graphite and can be imprinted with a design of the patient's choice. This feature allows for greater incorporation of the prosthesis into the patient's lifestyle and body image.

Ischial Containment Socket

The ischial containment socket is the most commonly used transfemoral socket design. The original design for this socket was described by Ivan Long, CP, as a "Normal Shape–Normal Alignment" design. This socket design takes a more anatomic approach to transfemoral socket fitting, with the ischial tuberosity now contained inside the proximal trim lines. The principle of this design is to provide a bony lock of the ischium in the prosthetic

socket. The inclusion of the ischium along with the lateral femur and pubic ramus prevents excessive abduction of the femur and increases medial-lateral stability during the stance phase of gait. This is especially helpful for individuals with shorter residual limbs and for those with mild hip abductor weakness. The distal two-thirds of this design have more muscle contouring compared with the quadrilateral socket. The critical design features of this socket design are described in the literature.⁶⁵

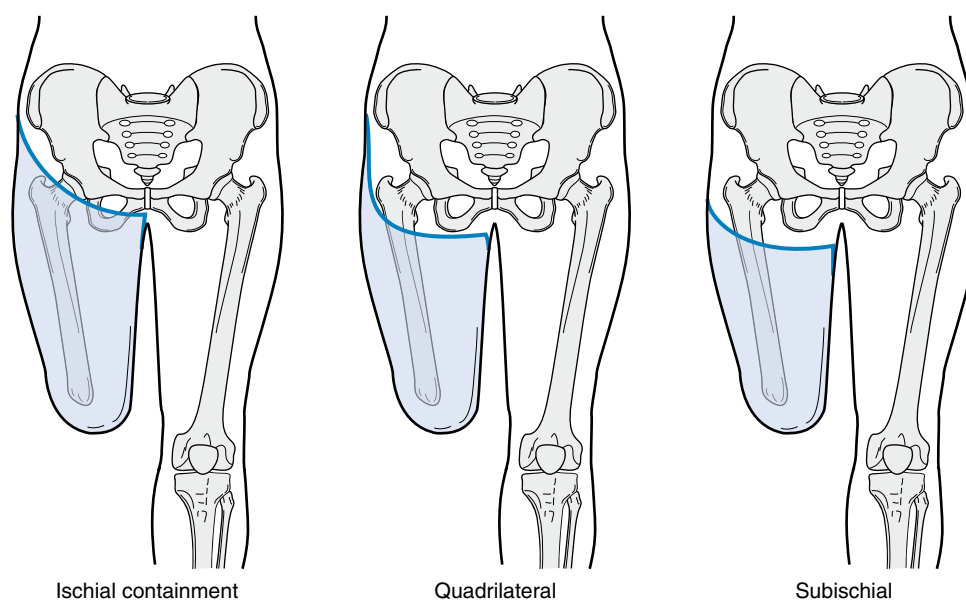
Subischial Socket Design

The second socket design is the subischial socket. The proximal trim line of this socket falls distal to the ischial tuberosity and relies completely on the thigh musculature for weight bearing.

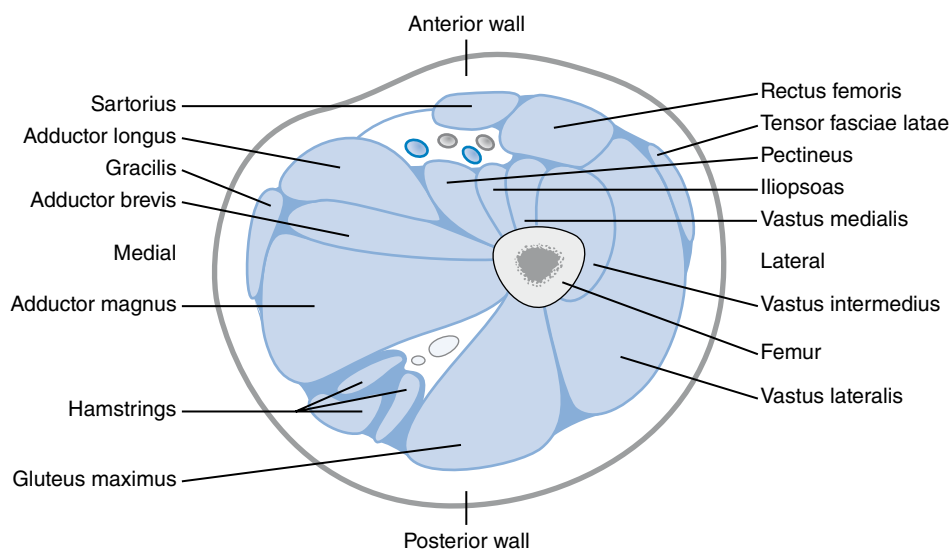
This socket design usually involves an elevated suction socket and requires a patient that is attentive to prosthetic care. Fig. 11.1 demonstrates the three different transfemoral socket designs.

Quadrilateral Socket (Quad Socket)

The quadrilateral socket type is rarely used today but can still be considered for limited ambulators who will sit in their prostheses for extended periods of time. As the name implies, this socket is rectangular with the medial-lateral dimension being greater than the anterior-posterior dimension (Fig. 11.2). The Scarpa's triangle provides a posterior-directed force, keeping the ischial tuberosity on a ledge, which is the major weight-bearing area of this design. The posterior shelf of this design provides the user with a stable



• **Fig. 11.1** Transfemoral socket designs. *Left*, Ischial containment. *Center*, Quadrilateral. *Right*, Subischial. (Based on an illustration by Brian Kausek.)



• **Fig. 11.2** Transverse cross section of the proximal aspect of thigh and quadrilateral socket. (From Schuch CM. Transfemoral amputation: prosthetic management. In: Bowker JH, Michael JW, eds. *Atlas of Limb Prosthetics: Surgical, Prosthetic, and Rehabilitation Principles*. 2nd ed. Mosby-Year Book; 1992.)

sitting surface. The distal two-thirds of the socket design have the general shape of the patient's residual limb and create total contact with little to no muscle contouring. There is some hydrostatic weight bearing through this more distal aspect of the socket.

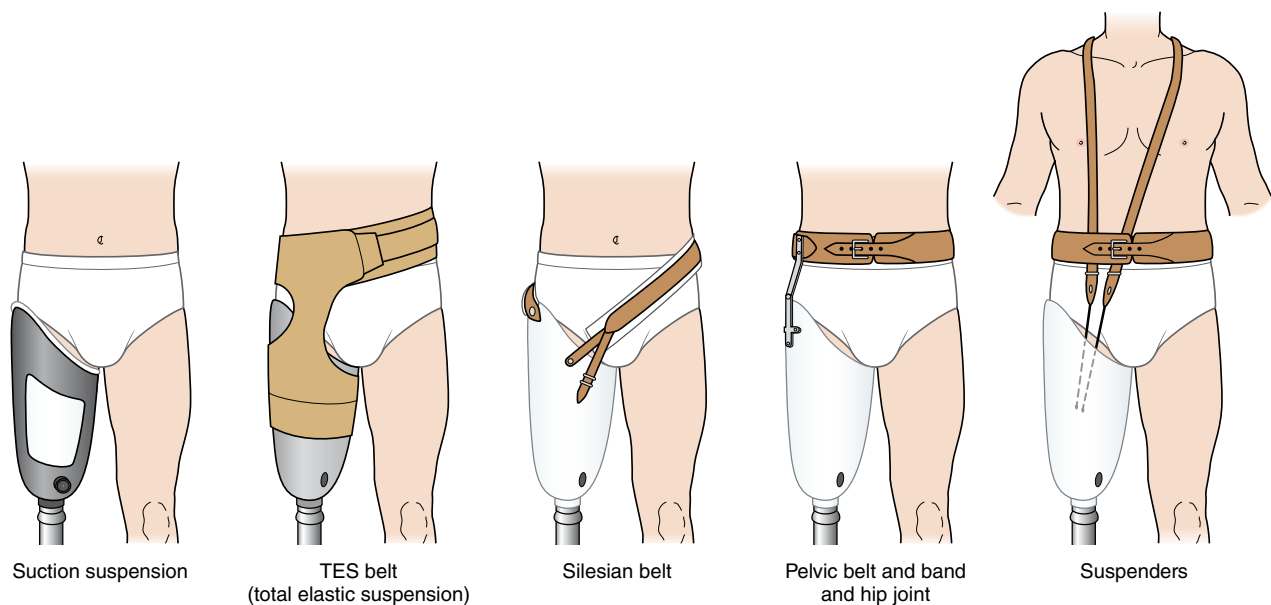
Prosthetic Limb Suspension

Suspension is the technique by which the prosthesis is connected and held on to a person's residual limb. Suspension can

be provided through a variety of mechanisms, including the anatomic shape of the limb, a liner, a sleeve, or with suction (Figs. 11.3 and 11.4). Suspension is a critical issue for successful use of lower extremity prostheses. Without proper suspension, the amputee patient cannot ambulate efficiently and will lack control over the prosthesis. Improper suspension can also increase the risk of falls with ambulation. This lack of suspension will negate the beneficial effects of any newer prosthetic component technology.



• Fig. 11.3 (A, B) Transfemoral socket suspensions.



• Fig. 11.4 Various transfemoral socket suspensions.



• **Fig. 11.5** (A) LimbLogic Electric Vacuum Pump System with key fob. (B) WillowWood One system. (Courtesy WillowWood, Mt. Sterling, OH.)

Suction Suspension

By creating a proximal seal within an airtight socket environment, suction sockets create a slight negative pressure to hold the prosthesis on to the residual limb. However, because of this negative pressure, a prosthesis with a lack of distal contact can create skin problems, including verrucous hyperplasia, if the lack of distal contact is chronic in nature. Historically, this approach has utilized a direct skin fit (no socks or liners between the residual limb and socket). The advent of gel liners has enhanced traditional suction suspension methods.

Elevated Vacuum Suspension

Elevated vacuum suspension can be seen as a derivative of suction suspension facilitated in part by the advent of interface liners (Figs. 11.5 and 11.6). In contrast to traditional suction socket systems, in which air is passively expelled from the socket through a one-way air valve, elevated vacuum systems actively draw additional air from within the socket environment. These designs provide a secure fit and stabilize the prosthesis firmly to the patient's residual limb, which can provide additional proprioceptive feedback and improved function, such as climbing a ladder or walking on uneven surfaces. There are currently several mechanisms that are used to create an elevated vacuum suspension environment, all of which require a TSB socket. Elevated vacuum systems maintain an enhanced negative pressure between the gel liner and socket wall. This negative pressure provides suspension and may assist with maintaining stability of the patient's residual limb volume. Elevated vacuum requires a pump that draws air out of the prosthetic socket. These pumps can be either manual or powered devices, and this technique can be utilized for either transtibial or transfemoral levels. Notably, elevated vacuum requires the maintenance of an appropriate socket fit and the user's attention to that fit. As such, it is not universally indicated.⁸³



• **Fig. 11.6** Ossur Unity mechanical vacuum pump incorporated into prosthetic foot. (Courtesy Össur, Reykjavik, Iceland.)

Pin Lock Suspension

Pin lock suspension requires a liner with a distal umbrella of an embedded mesh matrix and a threaded attachment site (Fig. 11.7). A locking pin can then be threaded into that liner, which will then engage into a locking mechanism attached to the



• **Fig. 11.7** WillowWood Alpha gel liner with pin, Pelite liner, and flexible inner socket.

prosthetic socket. The lock can be either a clutch mechanism or rotary style. Locks can be either push or pull to unlock. Pin lock suspension provides a secure mechanical link between the amputee and the prosthesis. The gel liner holds on to the patient, and the pin locks into the prosthesis. This system can be used for either transtibial or transfemoral amputation levels. Pin lock suspension can be very easy for the patient to use and offers an audible click to reinforce engagement of the lock. This is one of the simplest and safest designs. The biggest disadvantage is the consistent accurate donning of the liner and the potential for distal distraction and shearing forces on the tissues of the residual limb. The lanyard strap suspension is a version of pin lock suspension. Instead of a pin and locking mechanism, a Velcro strap is attached to the gel liner and fed through an opening in the distal portion of a prosthesis by the patient and fastened on the exterior portion of the socket. This system requires that the patient has some manual dexterity but may be simpler than the pin lock to manage. The lanyard strap is often favored in transfemoral applications because it reduces the build height distal to the prosthesis and facilitates a more proximal placement of the prosthetic knee joint. In addition, the lanyard allows the limb to be pulled, rather than pushed, into the socket, a favorable attribute for the additional soft tissues of the transfemoral limb.

Alternate Suspension Designs

There are many other methods of suspending a prosthesis. Boxes 11.3 and 11.4 list the most commonly used suspension methods for transtibial and transfemoral level amputations. Sleeves, straps, belts, and buckles have been used in the past and are still being used in some cases, either because of patient preference or because of anatomic considerations with the residual limb. These techniques can involve straps to proximal anatomic anchors. With transfemoral amputations, the suspension might involve a hip belt over the contralateral hip (Silesian belt) or a combination of a hip joint, pelvic band, and waist belt. The former may also be indicated for enhanced rotational control of the socket. In transtibial level amputations, this could include a cuff suspension strap that is just proximal to femoral condyles.

• BOX 11.3 Transtibial Suspension Systems

- Supracondylar and suprapatellar cuff/strap
- Supracondylar Pelite liner with compressible or removable wall
- Auxiliary suspension with sleeve
- Liner with pin-locking mechanism
- Suction with or without liner
- Vacuum
- Thigh corset and side joints

• BOX 11.4 Transfemoral Suspension Options

- Pin-locking or lanyard system
- Total elastic suspension belt
- Silesian belt
- Hip joint and waist belt
- Suction or partial suction (dry or wet fit)
- Suction with seal-in liner
- Vacuum

• BOX 11.5 Prosthetic Interface Options

- Pelite liner
- Urethane liners
- Thermogel/gel liners
- Silicone liners
- Special feature liners
- Hard interface directly with socket

Prosthetic Interface Options

The contact of the residual limb against the socket can utilize either a hard or a soft interface (Box 11.5). An example of a hard interface is found in a transfemoral suction socket in which the skin of the residual limb is in direct contact with the hard socket. In contrast, modern prosthetic liners provide protection against

both shear and impact forces and can be described as providing a soft interface. Soft interface options are indicated for and utilized by many amputees. Soft interfaces provide a cushion for the residual limb and allow for adjustment and comfort when the residual limb volume changes. Soft inserts are especially beneficial for patients with significant bony prominences or when invaginated, sensitive scar tissue is present. Soft interface materials include Pelite inserts, gel liners, and urethane or silicone liners. The disadvantages of soft inserts are their associated heat retention, susceptibility to wear and tear, added bulkiness, and tendency to absorb odors. Maintaining appropriate hygiene, as well as proper donning and doffing of these liners, can be problematic for some patients. Like all the other components of prostheses, there are many different brands and types of liners that are commercially available. Selection of the most appropriate interface system will depend on the individual needs and characteristics of the patient.

Prosthetic Feet

There are currently a wide variety of prosthetic feet that are commercially available. These feet are made of various materials and are produced by a host of manufacturers. Although each specific foot may have unique properties and specific advantages, this discussion primarily focuses on the different classes of prosthetic feet and the advantages and disadvantages of each class. There are a number of classification schemes that have been proposed for prosthetic feet. The classification system utilized in this text is included in Box 11.6. With advances in prosthetic designs, the features of one class can be combined with the features of another to produce a foot component that provides several different features.

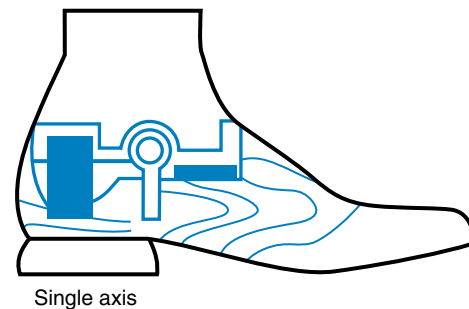
Nonarticulated Prosthetic Feet

Solid Ankle Cushion Heel Foot. The solid ankle cushion heel (SACH) prosthetic foot is a basic foot that is primarily indicated for individuals in the K1 functional level (Fig. 11.8). This foot has no moving parts, which makes it very lightweight, durable, and inexpensive. The foot simulates plantarflexion through the cushion heel portion of the foot. Because the keel of the foot provides limited movement, it offers no ability to effectively accommodate uneven surfaces.

Solid Ankle Flexible Endoskeletal Foot. The solid ankle flexible endoskeletal (SAFE) prosthetic foot is similar to the SACH foot in that it has no joint articulations and is durable and inexpensive. In contrast to the SACH foot, the SAFE foot allows some dorsiflexion, inversion, and eversion motion through the flexible



• **Fig. 11.8** Solid ankle cushion heel prosthetic foot. (Courtesy Ottobock, Minneapolis, MN.)



• **Fig. 11.9** Single-axis prosthetic foot.

keel of the foot. This provides greater ability to accommodate uneven terrain, but the motion is still limited and provides no energy-storing capacity. The SAFE foot is primarily indicated in K1 and low-level K2 ambulators.

Articulated Prosthetic Feet

Single-Axis Feet. Single-axis feet (Fig. 11.9) allow controlled motion at the ankle in dorsiflexion and plantarflexion. Rubber bumpers of various durometers control this motion and the ability of the foot to return to a neutral position. This allows the patient to achieve foot flat more quickly during the loading response phase of gait, which helps maintain stability of the prosthetic knee for individuals with transfemoral level amputations. Single-axis feet are typically indicated in individuals who are level K1 and K2 ambulators.

Multiaxial Feet. Multiaxial feet (Fig. 11.10) allow motion in multiple planes of movement, including axial rotation depending on the exact type of foot. Multiaxial motion can be obtained through the foot's flexible keel or through true mechanical joints. A flexible keel foot allows motion to occur within the keel itself as the ground reaction forces (GRFs) cause deformation of the foot. Multiaxial feet that allow motion through mechanical joints can allow motion in all three planes: plantarflexion and dorsiflexion, inversion and eversion, and transverse plane motion. Some multiaxial feet are fabricated with materials for energy storage and return as well. A disadvantage to this type of foot is the multiple moving

• BOX 11.6 Categories and Types of Prosthetic Feet

- Nonarticulated
 - Solid ankle cushion heel (SACH) foot
 - Solid ankle flexible endoskeleton (SAFE) foot
- Articulated
 - Single-axis foot: plantarflexion/dorsiflexion
 - Multiaxial foot
 - Hydraulic
- Energy storing/dynamic elastic response
 - Low profile
 - High profile
- Microprocessor control
- Microprocessor control with internal power
- Special activity feet



• **Fig. 11.10** Multi-axial prosthetic foot. (Courtesy Truststep, College Park Industries, Warren, MI.)



• **Fig. 11.11** High-profile energy-storing/dynamic response prosthetic foot with split keel. (Allpro feet, Courtesy Fillauer, Chattanooga, TN.)

parts, which may require more frequent repairs and maintenance. This type of foot is indicated for level K2 and K3 ambulators.

Energy Storing/Dynamic Response Feet. Energy storing/dynamic response feet (Fig. 11.11) are generally indicated for more active patients. These feet are made of materials (plastic or carbon fiber) that provide the capacity to store energy during weight bearing and then return energy once the foot is offloaded. The longer the spring, the more energy that is returned to the patient to provide push-off. Some energy storing feet also include a shock absorber mechanism to absorb vertical forces and reduce the GRFs that are transferred to a patient's residual limb. Depending on style, these feet can also provide similar motion as a multi-axial foot. This type of foot is indicated for level K3 and K4 ambulators.

Hydraulic Ankle Feet. Recent years have seen the development and acceptance of several hydraulic ankle feet in which hydraulic cylinders simulate both plantarflexion and dorsiflexion. This advancement has been especially valued during the negotiation of slopes and uneven terrain, and it appears to reduce internal socket forces during the negotiation of environmental obstacles and increased foot clearance in swing phase. Although initially developed as passive feet, subsequent models have integrated microprocessor control.

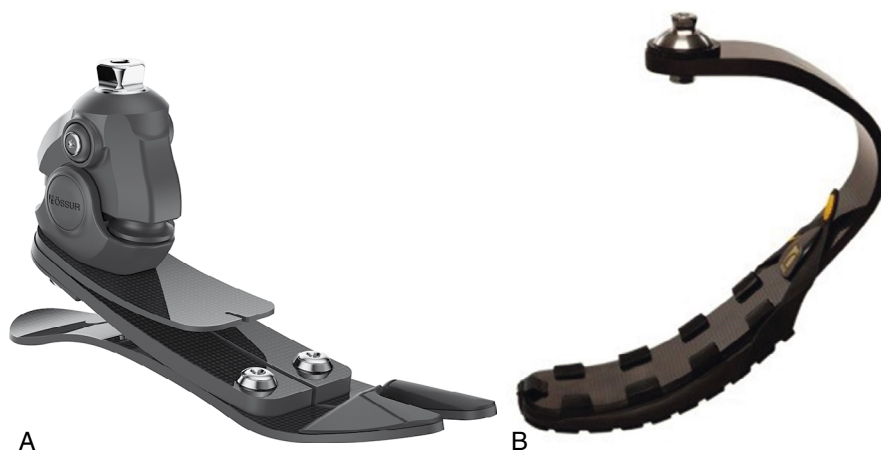
Microprocessor Feet. Microprocessor feet (Figs. 11.12 and 11.13) regulate the behavior of hydraulic ankle mechanisms and can be thought of as passive, active, nonpropulsive, or propulsive. Passive microprocessor feet feature microprocessors that



• **Fig. 11.12** Echelon prosthetic foot. (Courtesy Blatchford Products Limited, Basingstoke, UK.)



• **Fig. 11.13** Empower prosthetic foot with powered dorsiflexion and plantarflexion. (Courtesy Ottobock, Minneapolis, MN.)



• **Fig. 11.14** Össur adjustable-heel Pro-Flex LP Align (A) and Össur Running Foot (B). (Courtesy Össur, Reykjavik, Iceland.)

regulate the hydraulic resistance experienced at the ankle to accommodate a range of inclines, declines, and walking speeds. Active nonpropulsive microprocessor feet provide active power generation to produce ankle dorsiflexion during the swing phase of gait to assist with swing phase clearance. Propulsive microprocessor feet provide actively powered dorsiflexion and plantarflexion during both stance and swing phase. These can be especially useful for ambulating up inclines or hills, and there is some evidence to suggest that these devices reduce the energy cost of ambulation. Disadvantages to these feet are that they are heavy, cannot get wet, and, in some cases, need to be recharged at least daily. Current insurance coverage for microprocessor feet is inconsistent and varies with passive, active, and propulsive designs.

Specialty Feet. Feet in this category can accommodate various specialty needs (Fig. 11.14). For example, some feet have heel height adjustability to allow the amputee to wear shoes of different heel heights. These feet can vary heel height from a flat position to a heel height of 3 inches. Other examples of specialty feet are those designed to specifically accommodate running, rock climbing, and skiing.

Prosthetic Knees

Prosthetic knees can be classified according to the mechanical properties of the knee and the control system used for the facilitation or limitation of motion. They vary from very simple designs to sophisticated designs with microprocessor control. Selection and fitting of the most appropriate prosthetic knee should take into consideration multiple variables, including the activity level of the patient, residual limb length, and proximal muscle strength and motor control capabilities. The internal control mechanism for prosthetic knees can be as simple as mechanical friction, which provides constant resistance and works well for a single speed walker, to hydraulic fluid control with or without a microprocessor regulation to allow for varied gait speed and adjustability over variable terrain. As with the prosthetic foot prescription, prosthetic knees should be prescribed based on the patient's current functional needs and future goals and potential as well as on the environmental conditions during usage of the prosthesis. Box 11.7 lists the most common prosthetic knee design options.

• BOX 11.7 Prosthetic Knee Design Options

- Manual locking
- Single axis with constant friction
- Weight-activated stance control (safety knee)
- Polycentric
- Hydraulic or pneumatic swing phase control
- Hydraulic swing and stance control
- Microprocessor control (stance or stance and swing phase control)
- Microprocessor with internal power
- Hybrid units with combined features



• Fig. 11.15 Manual locking knee. (Courtesy Ottobock, Minneapolis, MN.)

Manual Locking Knee

Manually locking knees (Fig. 11.15) positively lock when fully extended and unlock by pulling a lever that is attached to the proximal portion of the socket (the lever is usually placed laterally or anteriorly). They cannot be flexed until unlocked. Manually locking knees provide the advantage of being the most stable knee designs, while the disadvantage of the knee is the compromised gait mechanics that occur because the patient must walk with a straight leg. This knee is utilized when stability is the primary issue for the patient, including during early bilateral fittings. This knee is relatively durable and inexpensive and is typically indicated for patients in the K1 functional classification category.

Single-Axis Knee

A single-axis, constant-friction knee is a basic knee joint similar to a hinge. It can have a spring-assisted extension so that the foot can advance more quickly during swing phase and achieve full knee extension sooner. The stability of the knee is based strictly on alignment and voluntary control, so patients need to have good proximal strength and control of the prosthesis to avoid knee instability. This knee is lightweight, durable, and inexpensive. It is typically indicated for patients in the K1 functional classification category.

Weight-Activated Stance: Control (Safety Knee)

A safety knee is a single-axis knee with a weight-activated locking mechanism. As the patient applies weight through the prosthesis during the stance phase, the knee locks automatically because of an internal braking system. For the braking mechanism to function appropriately, the knee cannot be flexed greater than approximately 20 degrees. When the patient's weight shifts off the prosthesis to initiate swing phase, the knee unlocks to allow flexion of the knee during swing phase. This knee can be used for new amputees and allows progressive adjustments from a very safe, locked status during stance phase to just a single-axis knee. This knee is generally indicated for amputees who fall into the functional classification K1 and K2 levels.

Polycentric Knees

A polycentric knee (Fig. 11.16) has inherent stability because of its construction and multiple points of rotation. These multiple joints create an instantaneous center of rotation that shifts as the knee flexes and extends. The instantaneous center of rotation is located more proximally and posteriorly with the knee in an extended position, which provides greater knee stability during the early part of stance phase. As the knee flexes, the instantaneous center of rotation shifts distally and anteriorly, which helps facilitate knee flexion in late stance phase. These knees are available with hydraulic or pneumatic swing phase control. Because of the relatively small distance between the knee axis of rotation and the attachment to the socket, this can also be an ideal knee for individuals with knee disarticulation level amputations. Such knees are often coupled with manual locking features to allow maximal stability in early gait training, maturing to enhanced mobility with gait training and experience.



• Fig. 11.16 Polycentric prosthetic knee with six-bar linkage. (Total Knee, Courtesy Össur, Reykjavik, Iceland.)



• **Fig. 11.17** Hydraulic swing and stance control knee unit. (Courtesy Össur, Reykjavik, Iceland.)

Hydraulic or Pneumatic Knees

Hydraulic or pneumatic knees (Fig. 11.17) include a cylinder that is either fluid (hydraulic) or air (pneumatic) filled that provides control during the swing phase of gait with knee extension and flexion. Some of these knee designs include mechanical features to provide stance control as well. These types of knees are appropriate for active patients that have good control and muscle strength. Because these systems provide the ability to adjust to faster or slower gait speeds, they are indicated for individuals who are able to ambulate with variable cadence (K3 functional classification level).

Microprocessor Knee

Microprocessor knees (MPKs) (Fig. 11.18) draw input from various types of sensors that provide input to the microprocessor. The types of sensors used include strain gauges, accelerometers, and gyroscopes, which provide detailed information on the knee angle, direction of movement, angular velocity, and weight-bearing status. The microprocessor analyzes this information at rates ranging from 50 to 1000 times per second to control the knee's resistance to flexion, increasing swing phase resistance when sensors indicate a misstep or stumble. These safety features facilitate the predominant benefits of such knee designs in reducing the incidence of stumbles and falls and reducing the cognitive load of ambulation. The MPKs have become more durable and reliable over time, but caution is still required with most of these knees when used around water or in other harsh environments. Although MPKs work well for many active patients, most are not designed to be used for running or other high activity sports. MPKs have historically been recommended for those rated at a functional index level K3 or K4; however, a growing body of evidence has identified many safety benefits with MPKs for higher-level K2 ambulators. There are some MPKs that provide only stance phase microprocessor control, which can be suitable for this population. Insurance companies often require good documentation of K level for reimbursement.



• **Fig. 11.18** Microprocessor knees. (A) C-Leg. (B) Waterproof X-3. (Courtesy Ottobock, Minneapolis, MN.)



• **Fig. 11.19** Microprocessor Power Knee, with internal power. (Courtesy Össur, Reykjavik, Iceland.)

Microprocessor Knees With Internal Power

There is currently one MPK that is commercially available that also has a motor capable of internal power generation (Fig. 11.19). This knee provides active knee flexion and extension, which is particularly useful with sit-to-stand activities and stair ascent. The disadvantages of this knee are the greater expense, heavier weight, and limited battery life. The knee is also relatively sensitive to environmental factors such as moisture.

Prosthetic Hip Joints

There are two primary styles of hip joints, single axis and multi-axial. The single-axis hip with an extension assist has been the most commonly used type of hip in the past. The multiaxial hip joint provides not only flexion and extension but also some rotation, which better simulates the normal motion of the hip. Use of a multiaxial hip joint coupled with an MPK is generally recommended for the person with a hip disarticulation level amputation who has a strong potential for prosthetic ambulation at the community level.

Additional Componentry Considerations

Sockets/rotation units. A multiaxial rotation unit allows for axial rotation in multiple planes. This is commonly called a torque absorber. It allows the foot to be firmly planted on the ground and the rest of the prosthesis to twist. Once the patient unweights the prosthesis, it automatically returns to its original position. This unit helps facilitate the twisting movements involved in playing golf and absorbs the rotational movements created during many daily activities. An endoskeletal axial rotation unit allows the patient to manually rotate the knee and foot to cross legs or tie shoes. This feature can be helpful in negotiation transfers in and out of a motor vehicle. An alignable system can be realigned during and after definitive fitting. *Lightweight* refers to materials such as titanium and carbon being used in the components of the prosthesis. A diagnostic or check socket is utilized by the prosthetist to ensure proper fit and function of the socket before fabricating the definitive socket. It is made from a clear plastic material, and the length of time that it can be worn depends on the type of plastic that is used.

Covers. Cosmetic covers consist of a type of foam (soft or hard) that is placed over the components of the prosthesis, making it look and feel like the person's contralateral limb. With the advent of new technology, carbon graphite components, and stylized socket designs, many patients prefer no cover. Moreover, when a cosmetic cover gets wet, it acts like a sponge. However, there are patients who need to see and feel something that looks like the leg that they lost. Prosthetic skins are a synthetic skin material made from silicone or other materials that match the person's skin tone. The two styles of prosthetic skins are custom and noncustom. Noncustom prosthetic skins are semidetached, and the patient and prosthetist pick a color from a swatch that closely resembles the patient's skin. Custom prosthetic skins are an exact match to the patient's contralateral limb and include hair patterns. These are extremely expensive and, in some cases, can cost as much as the actual prosthesis. Both styles add a measure of water resistance but are not fully waterproof for swimming. The prosthetic skin is susceptible to damage and can rip and tear secondary to insult. With a transfemoral prosthesis, a cosmetic skin that is continuous from foot to socket can impede the function of the prosthetic knee and make continued prosthetic adjustments challenging to perform. Prosthetic skins that protect a patient's cover from incontinence or extreme conditions need medical justification.

Prosthetic Prescription for Partial Foot Amputations

There are many levels of partial foot amputations ranging from a single ray to amputation at the level of the transtarsal joint with

sparing of the talus and calcaneus (Chopart amputation). One of the concerns with partial foot amputations is the distal end pressure and shear forces placed on the remaining foot. The prosthesis should be designed in such a way that these pressure points and shear forces are reduced. Use of a custom-fabricated, total-contact shoe insert can help to achieve this goal. The insert may include a filler to fill the space in the shoe left from the amputation. Utilization of a carbon fiber foot plate under the insert or use of rocker bottom shoe modifications can facilitate weight bearing and allow for a more normal gait pattern. For proximal partial foot amputations in the midfoot or hindfoot, incorporation of an ankle-foot orthosis should be considered to enhance the stability of the foot and ankle complex and allow greater functional activity levels. The patient's footwear should also be taken into consideration during the prescription.

Prosthetic Prescription for Ankle Disarticulation (Syme) Amputations

With a disarticulation amputation at the level of the ankle joint (Syme level), the heel pad is kept in place to create a weight-bearing surface. One of the advantages of this surgery is to allow the patient to take limited steps without a prosthesis. However, one disadvantage is that the length of the residual limb precludes the use of most energy-storing feet. In addition, the heel pad can migrate off the distal end over time and render the distal end nonweight bearing. Because of the bulbous anatomic shape of the distal end, suspension is typically not a problem. However, the bulbous end can create cosmetic problems with the prosthesis. The socket design, specifically the height of proximal trim line, depends on the distal weight-bearing capacity. If a patient has no weight-bearing ability, then the trim line will need to be brought up to the patellar tendon level. The proximal trim line will move distally to the tibial tubercle if the heel pad is in place and tolerates weight bearing. The liner can be a custom gel liner or use traditional padding, such as Pelite. The suspension can be built into the liner (extra gel or padding above malleoli) or can be a window and door configuration medially or posteriorly. Prosthetic foot options include a low profile Syme, SACH foot, or carbon composite foot. The carbon foot allows some energy storing and provides better accommodation over uneven surfaces. A stable alignment at this amputation level requires a relative outset of the prosthetic foot that further compromises the appearance of the prosthesis.

Prosthetic Prescription Algorithm: Transtibial

The choice of components for the transtibial prosthesis prescription depends on the individual's current or potential functional abilities and the patient's goals for prosthetic use. The following recommendations for each functional level are for consideration, and it is important to develop a prescription that is individualized. The essential elements for a transtibial prosthesis that need to be included for every functional level include a socket, interface, suspension, pylon/frame, and type of foot and ankle (Fig. 11.20). Items included in all prescriptions for a prosthesis are a clear diagnostic socket and socks (single and multi-ply, 6 each). The patient can select a custom-shaped cover and/or prosthetic skin to help address cosmetic concerns.



• **Fig. 11.20** Transtibial prosthesis. (Courtesy Ottobock, Minneapolis, MN.)

Functional Level One (K1)

Patients in this functional category have the ability to use a prosthesis for transfers or to ambulate over level surfaces for short, household distances. Safety is the greatest priority for this population. The socket design should be total contact style with special considerations for comfort during sitting. The type of interface and suspension system used should take into consideration the patient's ability to don and doff the prosthesis and manage personal hygiene independently. Recommended foot and ankle components are the nonarticulated feet, such as the SACH or SAFE foot, or a simple articulated foot, such as the single-axis foot. Also included in this prescription will be a clear diagnostic socket, prosthetic socks, and a cosmetic cover.

Functional Level Two (K2)

Patients in this functional category have the ability to perform limited community distance ambulation and traverse some environmental barriers. The major changes in the prosthetic prescription will be that the components should be alignable, and the prosthetic foot can be a multiaxial or a flexible keel-type foot to allow for accommodation over uneven terrain. Suspension for this group can utilize a pin lock, sleeve suspension, or suction suspension with a sleeve and one-way expulsion valve in the socket.

Functional Level Three (K3)

Functional level K3 amputees are community distance ambulators who have the ability to traverse most environmental barriers and ambulate with variable cadence. Special considerations

for this group will be the type of prosthetic foot. This will be some type of energy-storing (dynamic response) foot, and, depending on the activities that they are doing, it can include a dynamic pylon or features that allow greater accommodation over uneven terrain. Foot and ankle components that incorporate hydraulic units with or without microprocessor control also can be considered in this population. A foot and ankle component with both microprocessor control and internal power may also be indicated in this class of patients. An additional consideration for prosthetic suspension is the use of an elevated vacuum technology.

Functional Level Four (K4)

Patients in this classification level have the ability or potential ability for ambulation that exceeds normal requirements. This may include sports or recreational activities that require high impact, stress, or energy levels that are typical of the prosthetic demands of a child, high-activity adult, or athlete. At this level, specialty components such as running feet can be prescribed. Suspension is also key for this group to avoid catastrophic failure of the prosthetic connection during the activity. This may include use of a backup or secondary suspension method. Special considerations are also needed for the pediatric population because of growth and the deterioration of components secondary to the high utilization.

Prosthetic Prescription Algorithm: Knee Disarticulation

A knee disarticulation level amputation leaves the femur intact and creates a distal weight-bearing surface with retained thigh musculature. This long lever provides for potentially better control of the prosthetic limb and maintains the distal growth plate of the femur, which is important for individuals who are skeletally immature at the time of amputation. The disadvantages to this level of amputation are the discrepancy between the height of the prosthetic knee center and the height of the contralateral anatomic knee as well as the decreased cosmetic appearance. The prosthesis proximal trim line will depend on the patient's ability to bear weight on the distal end of the residual limb. If true full distal weight bearing can be tolerated, then the proximal trim can be lowered to the subischial level. If there is no distal end weight-bearing capacity, then the residual limb is treated as a transfemoral amputation with use of a more traditional ischial containment socket.

Prescription Criteria

Socket design for this level typically includes an anatomically shaped socket with a flexible inner socket. The proximal trim lines of the socket will be determined based on distal weight-bearing tolerance as noted previously. Interface and suspension options are generally the same as those for the transfemoral level, but there is also the possibility of creating self-suspension of the prosthesis with use of the femoral condyles. Polycentric knee units are commonly recommended to reduce the difference in knee centers from the prosthetic limb side to the intact limb side. Depending on the functional goals of the patient, additional features, such as a hydraulic mechanism, may be indicated and beneficial.

Prosthetic Prescription Algorithms for Transfemoral Amputees

The transfemoral prosthesis prescription is also based on the individual's current or potential functional abilities as well as the goals for prosthetic use. In addition to the components included in the transtibial prosthesis prescription, the transfemoral prescription will include the prosthetic knee. Other unique considerations for the prosthetic prescription at the transfemoral level include the greater need for safety and stability compared with the transtibial level. The provider should also take into consideration the greater metabolic cost of ambulation with a transfemoral prosthesis, the greater challenges associated with donning and doffing, and the concerns regarding sitting comfort with a transfemoral prosthesis (Fig. 11.21).

Functional Level One (K1)

When evaluating the person in this functional category, it should be remembered that the goal is limited household walking on level surfaces. Primary prosthetic considerations are for safety and ease of prosthetic use. During stance phase, the knee needs to remain stable or locked, so the use of knee components, such as a weight-activated stance control knee or manual locking knee, are appropriate. Pairing a manual lock with a polycentric knee can facilitate transition to a more efficient gait with training and experience. Choosing a prosthetic foot, such as a SACH foot with a soft heel, or a single-axis foot can also assist in maintaining knee stability by allowing or simulating ankle plantarflexion during the loading response phase of gait. This helps keep the GRF anterior



• **Fig. 11.21** Transfemoral prosthesis. (Courtesy Ottobock, Minneapolis, MN.)

to the knee center, thus creating a knee extension moment. The type of socket can be an ischial containment or quadrilateral style and should have a flexible inner liner and a flexible proximal brim with cutouts in the posterior portion to allow for sitting comfort. The socket material can either be a laminate or plastic. A gel liner with pin lock or lanyard can be used for suspension, recognizing the impact of pin-lock mechanisms in lowering the prosthetic knee center. A hip joint and waist belt method of suspension can be used if there are concerns about safety or if needed from a stability standpoint.

Functional Level Two (K2)

Transfemoral amputees who fall in this functional category still typically benefit from knee components that provide enhanced levels of stance phase stability while allowing greater swing phase motion and function for longer distances and a more normal gait pattern. Examples of prosthetic knees that provide these features include weight-activated, stance control knees and polycentric knee units. A growing body of evidence continues to suggest that some high-level K2 ambulators may benefit from knee units that offer microprocessor stance phase control. Knee units that offer hydraulic or pneumatic swing phase control are not indicated for the level K2 transfemoral amputee because these individuals do not have the ability or potential to ambulate with variable cadence. Components should be adjustable for alignment. The prosthetic foot can be a multiaxial type or a flexible keel style for better accommodation over uneven terrain.

Functional Level Three (K3)

Patients in this category should have the ability to perform full community ambulation and ambulate with variable cadence. The type of socket selected can be ischial containment or subischial. The socket should have a flexible inner liner with cutouts in the posterior portion to allow for sitting comfort and a flexible proximal brim. In addition to all the previously described choices for suspension, elevated vacuum is a consideration at this level. Although stability and safety are still important, these individuals also require prosthetic knees that provide enhanced levels of function. Knee units with pneumatic or hydraulic swing phase control allow for variable cadence ambulation. Polycentric and microprocessor features provide for positive stance stability and a more natural and symmetric gait pattern. An internally powered knee may be indicated in circumstances where stair and incline ascent are essential. At this level, multiaxial feet and energy storing feet, with or without a dynamic pylon or hydraulic ankle, can be considered.

Functional Level Four (K4)

Patients who achieve this level of classification have the ability to use their prosthesis for activities outside of normal ambulation. Because of this, they may benefit from a prosthesis that is used for their regular day-to-day activities as well as a prosthesis that is used for participation in specific athletic or recreational activities (Fig. 11.22). The need for a specialty prosthesis may also depend on the frequency and intensity of the recreational or sport activity. If an entirely separate prosthesis is not required for participation in the desired activity, the patient may benefit from components with features that can serve multiple purposes. There is also the possibility of using a coupler or other quick disconnection devices



• **Fig. 11.22** Specialty transfemoral prosthesis designed for running activities. (Fitness Prosthesis, Courtesy Ottobock, Minneapolis, MN.)

that allow for the exchange of specific components for different activities. There are many varieties of prosthetic components to allow the amputee to participate in a full range of activities. For example, some components facilitate running, and others can help the amputee to participate in scuba diving. Socket fit and suspension are particularly critical for these types of prostheses. The environment in which the prosthesis will be used for the desired activity also needs to be taken into consideration. In general, this requires the use of components that are relatively simple and durable; however, some advanced MPKs have now been designed to allow running, full submersion in water, and performance in various environments.

Hip Disarticulation and Hemipelvectomy Levels of Amputation

There are several challenges associated with prosthetic fitting and use with these more proximal levels of amputation, including the need to replace three anatomic joints and the significant increase in metabolic cost of ambulation. Because of the lack of a residual limb with these amputation levels, socket fit, prosthesis suspension, and alignment are also major considerations. Although newer lightweight components are available, the weight of the prosthesis is also a factor (Fig. 11.23).

Prescription Criteria

Once called a “bucket socket,” the socket at these amputation levels has historically been total contact, extending superiorly over

the ipsilateral iliac crest into the waistline and to the level of the thoracic spine for hemipelvectomy-level amputations. More recent socket variations have explored reducing this surface area. It is essential to capture all motion inside this socket because the patient utilizes trunk flexion and extension to produce and control motion in the prosthesis. The patient’s functional ability and goals help to determine whether a single-axis or multiaxial hip joint is used. For the knee joint, either a weight-activated stance control with extension assist or an MPK is typically recommended to provide enhanced stance phase stability of the prosthesis. Prosthetic foot recommendations include single-axis feet for those who will primarily use the prosthesis over level surfaces to multiaxial or dynamic response feet for those with the potential for more community ambulation. To satisfy cosmetic needs, a custom-shaped cover or even a prosthetic skin that protects patient’s cover from incontinence or extreme conditions can be added. Because of the high energy requirements associated with gait and sitting discomfort, individuals with this level of amputation may not use the prosthesis regularly, preferring the simplicity of crutch-assisted ambulation.

Alternative Socket Designs

The field of prosthetics continues to search for the perfect socket design, one that allows the lower extremity amputee to function at a high level yet be extremely comfortable. Many of these socket designs include adjustable panels in the rigid frame that can either be tightened or loosened as the person’s residual limb volume changes (Fig. 11.24). Transtibial designs include iFIT socket, which is a noncustom design and can be used as a preparatory



• **Fig. 11.23** Hip disarticulation prosthesis. (Courtesy Ottobock, Minneapolis, MN.)



• Fig. 11.24 (A–D) Specialty socket designs.

prosthesis. The REVO socket design is a technique that adds patient-adjustable panels to a custom socket. Some transfemoral socket designs, such as Martin Socket-less socket and Limb Innovation's Infinite socket, have very little socket other than struts and proximal brim. There is also the Bikini socket by Martin prosthetics for individuals with hip disarticulation-level amputations. All of these designs are a modern twist on older designs utilizing new materials and techniques (Fig. 11.25).

Although their introduction to the United States has been slowed by regulatory requirements, percutaneous osseointegration (OI) implants have been developed and utilized to achieve direct skeletal attachment of a prosthetic limb to the residual limb of a person with an amputation. Compared with socket suspension techniques, direct skeletal attachment of a prosthetic limb through OI offers many potential advantages, including improved mechanical transfer of motion, reduced skin irritation from a prosthetic socket, improved joint ROM, and enhanced comfort. However, OI also presents the possibility of serious risks, such as



• Fig. 11.25 Historical evolution of prostheses. (Courtesy ABC Local, Clare Rawlinson, Australia.)

infection, failure at the bone-implant interface, or skeletal fracture. Placement of an OI implant may also require a longer period of postoperative recovery and rehabilitation compared with traditional amputation. Placement of these implants is currently viewed as contraindicated in those with significant peripheral vascular disease or diabetes.⁵¹

Currently, the US Food and Drug Administration (FDA) considers OI implants to be Class III medical devices, requiring the highest degree of regulatory oversight to ensure that the devices are safe and effective. Several different types of percutaneous OI implants are currently being used for the direct skeletal attachment of prosthetic limbs in the United States, either under full market FDA approval, research protocols, through FDA-approved compassionate use exemptions, or as custom implants. There are also OI implants in use outside the United States, such as the Badal X (OTN Implants) and the osseointegrated prosthetic limb, that have not been approved by the FDA for implantation within the United States. Although this rapidly evolving field holds promise to enhance outcomes for individuals with limb loss in the future, it poses many regulatory uncertainties with respect to immediate and long-term prosthetic fittings.

Prosthetic Limb Fitting and Replacement Considerations

The length of time from amputation surgery to the clinic visit for prescription of a prosthesis will determine what type of prosthesis is utilized. With an initial prosthesis, basic components can be utilized to allow for a quicker fitting. However, with current component availability, it is usually better to prescribe the definitive prosthesis and have the prosthetist perform a socket replacement when the patient has a significant anatomic change to the residual limb.

A prosthesis will typically last for 3 to 5 years, depending on the activity of the patient and anatomic changes in the residual limb. However, supplies such as interface liners and socks require more frequent replacement. These usually need to be ordered every 6 months to 1 year. A change in the residual limb (reduction or increase in size), surgical revision of the residual limb, pediatric and adolescent growth, and broken components all may require a new prosthetic prescription. The age of the prosthesis and how well the patient is doing with the prosthetic components will determine whether an entirely new prosthesis is required or whether only a socket replacement is needed. The other factor for consideration of a socket replacement vs. providing a new prosthesis will be insurance guidelines and reimbursement. The typical wearing schedule for a new amputee is outlined in Table 11.3. The patient should follow skin care guidelines and do frequent skin checks. Follow-up appointments should be made for the patient to see a physician and prosthetist during week 1.

Energy Consumption

When adults choose a walking speed that is comfortable and customary to them, their rate of oxygen consumption is fairly consistent and not significantly different during aging up to the age of 80 years.^{6,61,71} When ambulating with a prosthesis, the customary walking speed (CWS) is typically slower but generates an oxygen consumption rate that closely mirrors that of individuals without amputation or pathology affecting their gait. However, the amputee's gait is less efficient than an individual

TABLE 11.3 Typical Wearing Schedule for New Amputee

Day	Morning	Afternoon
Week 1		
1	1 h (20 min on, 20 min off)	0
2	1.5 h (20 min on, 20 min off)	0
3	1 h	1 h
4	1.5 h	1.5 h
5	1.5 h	1.5 h
6	2 h	1.5 h
7	2 h	2 h
Week 2		
8	2.5 h	2 h
9	2.5 h	2.5 h
10	2.5 h	2.5 h
11	3 h	2.5 h
12	3 h	3 h
13	3 h	3 h
14	3.5 h	3 h
Increase time to full time.		

without an amputation, which results in greater levels of energy consumption for traveling a given distance (metabolic cost). The metabolic cost of ambulation increases with more proximal levels of amputation. Thus the increase in energy consumption for a given distance (metabolic cost) for a traumatic transtibial or transfemoral amputee walking with a prosthesis at a comfortable walking speed is around 25% and 63%, respectively (these figures can vary across studies somewhat). Increases in metabolic costs for dysvascular amputees at the same amputation levels are 40% and 120%, respectively.⁶ Standard walking speeds are generally established for use in research comparison studies to self-selected walking speeds, which represent the walking speed a given individual can comfortably maintain. In the cases of individuals with amputation, many factors influence the self-selected walking speed. Some of these include:

- 1. Laterality (unilateral vs. bilateral)
- 2. Level of amputation
- 3. Comorbidities and the health of various organ systems (e.g., pulmonary and cardiovascular systems)
- 4. Cognition
- 5. Prosthetic componentry

Interestingly, ambulation with crutches with a swing through gait and without a prosthesis shows higher values for energy expenditure compared with prosthetic ambulation. Ambulation with a prosthesis thus shows improved energy conservation for these individuals. In general, dysvascular amputees have a slower CWS and less efficient gait than traumatic amputees with comparable amputation levels. In studies of residual limb length in transtibial amputees with respect to these parameters, no correlation could be found. However, transtibial amputees with short residual limbs showed a faster CWS and a lower oxygen cost than

amputees with knee disarticulations or transfemoral amputations. This finding underscores the importance of preserving a knee joint if possible. For bilateral amputees at a given level, energy expenditure is greater than for unilateral amputees at that level. However, energy consumption for bilateral transtibial amputees is less than that for unilateral transfemoral amputees. Again, a distinction is found between dysvascular and traumatic bilateral amputees. The latter walk faster and more efficiently than their counterparts with comparable amputation levels.

Until recently, most studies on energy consumption and gait in amputees have been done on amputees using components with passive systems. In the emerging era of powered knees and foot/ankles, gait parameters can change. For example, an improvement of 10% in self-selected walking speed occurred when a powered foot/ankle was used compared with one without power.³⁵ In another study, the energy cost of walking was reduced when a bionic foot/ankle was used compared with walking with a dynamic carbon fiber foot.²¹

Bilateral Amputee Considerations

Amputees affected on both limbs, such as bilateral transfemoral amputations, bilateral transtibial amputations, or transfemoral amputation on one side and a transtibial amputation on the other side, face additional challenges. Bilateral amputations can be a result of disease processes, such as peripheral vascular disease or traumatic injury. Studies have shown that up to 50% of unilateral dysvascular amputees will become bilateral amputees over a 5-year period. Trauma from transportation or industrial accidents, electrocution, and war-related incidents, such as roadside bombs or land mines, are generally responsible for bilateral amputations that occur simultaneously.

Success as a unilateral amputee can be a helpful predictor of success in ambulation as a bilateral amputee. The metabolic cost of ambulation increases during prosthetic ambulation with bilateral amputations. The more proximal the amputations, the higher the associated energy cost. Energy requirements increase for unilateral amputees at a certain level and then increase further if there is another amputation on the contralateral limb at the same level. Because of these increased needs, there must be sufficient cardiac capacity and strength capacity to bear these extra burdens. Flexion contractures of the hips or knees can restrict or inhibit prosthetic ambulation in amputees in general but pose even greater issues for the bilateral amputee. The early intervention of the rehabilitation team is therefore paramount in the prevention of contractures.

In general, training with bilateral prostheses requires more time and effort. Because there is loss of proprioceptive sensation from both lower extremities with bilateral amputation, use of bilateral prostheses can cause a greater sense of insecurity that may result in a wider stance, slower pace, and use of an assistive device. The bilateral amputee faces a greater challenge when negotiating ramps, curbs, stairs, uneven terrain, and other environmental barriers. Important skills would also include sitting and standing from a chair, falling in a controlled manner, and recovering from a fall.

In bilateral amputation, preservation of residual limb length can have a dramatic impact on the success of prosthetic ambulation. This is especially true if the knee can be preserved.⁶³ The proportion of amputees with bilateral transfemoral amputations secondary to peripheral vascular disease who will successfully ambulate with bilateral prostheses is small. If the bilateral trans-

femoral amputations are trauma related and peripheral vascular disease is not present, then successful ambulation is more likely. Preservation of one or both knee joints will also contribute to successful ambulation. Individuals with unilateral below-knee amputations have a higher likelihood of ambulating than those with bilateral below-knee amputation. Many factors can affect outcome, such as body mass index, cognition, comorbidities, preamputation ambulatory status, and whether the amputation was caused by trauma or vascular disease. One study⁷⁰ found that 12.5% of individuals who had bilateral below-knee amputations caused by severe ischemia were able to “walk out of home.”

The same pool of components used for unilateral amputees are used for bilateral amputees. Most of the same principles for selection apply. In a majority of cases, the same ankle-foot complex should be used on both sides. However, this is not an absolute rule, and a given individual may have requirements that cause a deviation from this guideline.

Stubbies can provide a useful initial phase in the training of bilateral transfemoral amputees to ambulate. Stubbies consist of an ischial containment socket, a pylon, and specially designed feet to give more stability when leaning backward. The pylons are adjusted to a length that facilitates sitting down in chairs. The feet are of two types. One version is triangular. The pylon inserts at the apex, which is anterior, and thus the base is posterior. The other version is a bean shape with the pylon attaching at one-third of the longitudinal distance with the longer arm of the foot positioned posteriorly. Once the amputee has acquired the necessary core strength and balance along with competence in ambulation, the stubbies are advanced by increasing the pylon length. Locking knees are frequently indicated with extended pylon lengths to facilitate sitting in confined areas. Progression to standard prosthetic feet and MPKs will follow for those individuals with adequate physical strength and balance, interest, and motivation. Notably, the restoration of preamputation height is not necessary, with many bilateral prosthesis users preferring shorter prostheses.

Pediatric Lower Limb Loss

The epidemiology and outcomes of limb absence in the pediatric population are very different when compared with the adult population. In this population, congenital limb deficiencies outnumber acquired amputations by more than 2 to 1.^{74,92} In contrast to the older adult population, where prosthetic usage and benefit are sometimes uncertain, usage rates of lower limb prostheses in the pediatric amputee population are very high.^{5,92} A study of 258 children with limb deficiencies between the ages of 2 and 21 years suggests that prostheses are often worn for more than 12 hours a day and are commonly used in activities, such as swimming, running, bicycling, and basketball.⁹² Similar findings with regard to cycling, swimming, and sports activities have been reported elsewhere.⁵ Health-related quality of life for children and adolescents with limb reduction deficiencies has been noted to be higher than that observed in a cohort of their peers with other chronic health conditions, including asthma, arthritis, diabetes, cystic fibrosis, cerebral palsy, and epilepsy.¹⁰⁰ Findings are similar to able-bodied peers, as well.⁶⁰ This last statement appears to be unaffected by the degree of lower limb deficiency, the presence of bilateral lower limb deficiency, or a concomitant upper limb deficiency. Only a slight decrease in both the diversity and frequency of social and skill-based activities among adolescents with lower limb deficiencies has been noted.⁶⁰

Prevalence, Classification, and Epidemiology: Congenital Lower Limb Deficiencies

The prevalence of congenital limb defects has been reported at 21 cases per 10,000 births.⁹³ However, figures on limb defects include the prevalence rates of both syndactyly and polydactyly as well as upper limb reduction deficiencies, which are generally reported to be at least twice as common as lower limb reduction deficiencies (LLRDs). The prevalence of LLRD is much lower, with recent observations suggesting 1.6 to 3.0 cases per 10,000 live births in numerous global datasets.^{2,49,86,93} Bilateral LLRD appears to be common, reported among 22% to 37% of those with LLRDs,^{3,66} similarly, upper limb reduction deficiencies have been reported in 30% to 40% of those with LLRD.^{2,74,87} In addition, roughly half of those children with congenital lower limb deficiencies present with other major anomalies, including anomalies of the internal organs, axial skeleton, and central nervous system, reported in 26%, 14%, and 13%, respectively, of those with congenital lower limb deficiencies.⁸⁷

Several different mechanisms have been developed for the classification of LLRDs. Although formally replaced, the traditional classification system of Frantz and O’Rahilly is still in use in some practice environments, and an overview of this classification system is included in Table 11.4. Within this system, LLRDs are defined as either “terminal,” with a complete loss of the distal extremity, or “intercalary,” in which an intermediate portion of the limb is deficient with intact elements of the limb both proximal and distal to the deficiency. Terminal LLRDs include both “transverse,” in which there is complete absence of the limb at the level of the loss, and “paraxial,” in which there is complete longitudinal absence of either the preaxial element (tibia) or postaxial element (fibula). Intercalary deficiencies include both “paraxial” events, suggesting the localized absence of elements of the tibia or fibula with preserved proximal and distal elements, and “phocomelia,” in which there is an absence of central elements of the femur with foreshortening of the otherwise preserved lower limb. Within this structure, the terms *amelia* and *hemimelia* are also used to denote the complete or partial absence of the limb, respectively. Terms such as *fibular hemimelia* and *proximal femoral phocomelia* have persisted despite the formal replacement of this classification system.

The current standard in the classification of LLRDs was introduced in 1989 by the ISO and identified as ISO 8545-1, Method of Describing Limb Deficiencies Present at Birth^{45a} (Table 11.5). Within the ISO classification, LLRDs are categorized as either “transverse” or “longitudinal.” Transverse deficiencies suggest the absence of any skeletal elements distal to a certain level of the limb. The full description of a transverse deficiency requires the

naming of the affected limb segment and the approximate location. Thus transverse deficiencies of the thigh or leg should be qualified as being “complete,” “upper third,” “middle third,” or “lower third.”

According to the ISO system, any LLRD that is not “transverse” is considered “longitudinal,” suggesting a reduction or absence of an element within the long axis of the limb. The naming of such deficiencies is done from proximal to distal, listing the bones or segments and whether these segments are totally or partially absent. Within this nomenclature, a hypothetical deficiency formally described as paraxial tibial hemimelia might be described as “longitudinal deficiencies of the tibia total, tarsus partial, ray 1 total.” Similarly, proximal femoral focal deficiencies, or PFFDs as they have long been called, are now formally described as “longitudinal deficiencies of the femur, partial.”

More recently, a 2011 publication by the Active Malformations Surveillance Program at the Brigham and Women’s Hospital in Boston, Massachusetts, suggested that the full classification of congenital limb deficiencies should also include the apparent etiology.³⁷ These include the categories of chromosomal abnormalities, known syndromes, Mendelian inheritance, familial inheritance, presumed vascular disruption defects, teratogenic exposure, and unknown causes.

Classification and Etiology: Acquired Amputation

Pediatric-acquired amputations should be described with the same ISO system and classification system that is used in the adult population with acquired limb loss. (Refer to Table 11.1 for the details of this classification system.) The leading cause of acquired amputation in the pediatric population is trauma, with US national statistics estimating roughly 1000 new cases each year, with a 3:1 male to female ratio.¹⁷ Within this category, the most commonly encountered mechanisms of injury are those in which a body part is caught in or between objects (18%), followed by machinery (17%), lawn mower injuries (12%), and motor vehicle accidents (8%).⁷⁸ Finger and thumb amputations (54%) and toe amputations (20%) are the most common locations for pediatric amputations, with transtibial and transfemoral amputations accounting for only 15% of traumatic acquired amputations in the pediatric population.⁷ These major amputations are more common in older patients with 68% of transtibial amputations occurring in children 12 years old or older and 49% of transfemoral amputations occurring in 16- to 17-year-olds.

With respect to acquired lower limb pediatric amputations, 75% of the amputations associated with lawn mower injuries involve the lower extremity. The vast majority occur in younger children

TABLE
11.4

Frantz and O’Rahilly Traditional Classification System of Congenital Skeletal Lower Leg Deficiencies

TERMINAL DEFICIENCIES		INTERCALARY DEFICIENCIES	
No unaffected parts distal to the deficient portion		Immediate parts are deficient with proximal and distal elements present	
Transverse	Paraxial	Paraxial	Phocomelia
Complete absence distal to the level of loss	Complete longitudinal absence	Absence of preaxial or postaxial elements	Absence of central elements with foreshortening of the limb
Amelia Hemimelia	Hemimelia (tibial or fibular)	Hemimelia (tibial or fibular)	Phocomelia (femoral)

TABLE 11.5 International Organization for Standardization Classification System of Congenital Skeletal Lower Limb Deficiencies

Transverse Deficiencies	Longitudinal Deficiencies	
No unaffected parts distal to the deficient portion	Any deficiency that is not transverse, named from proximal to distal	
Thigh	Femur	
Complete	Total	
Upper third	Partial	
Middle third		
Lower third		
Leg	Tibia	Fibula
Complete	Total	Total
Upper third	Partial	Partial
Middle third		
Lower third		
Tarsal	Tarsus	
Complete	Total	
Partial	Partial	
Metatarsal	Metatarsals (1–5)	
Complete	Total	
Partial	Partial	
Phalangeal	Phalanges (1–5)	
Complete	Total	
Partial	Partial	

(57% in 0- to 5-year-olds, 20% in 6- to 11-year-olds), accounting for 50% of all transtibial amputations, and 39% of all transfemoral amputations were observed in children 0 to 5 years old.⁷

By contrast, motor vehicle accidents were the leading cause of adolescent amputations, with more than half occurring in patients of driving age (16–17 years old) and accounting for 35% of transtibial amputations and 36% of transfemoral amputations observed in children over the age of 12 years. The most common amputation locations in motor vehicle accidents are fingers (32%) followed by transtibial (24%) and transfemoral amputations (14%). Motor vehicle accidents generally cause more severe amputations, with 82% of lower extremity amputations occurring proximal to the toes.⁷ Children living in poverty, those identified as racial or ethnic minorities, or those living in single-parent homes also appear to be at elevated risks for amputation as a result of injury.⁹⁷

With respect to disease, malignant tumor remains the most common cause of amputation, accounting for over half of these events, with the highest incidence occurring in the second decade of life.⁹⁷ Commonly encountered tumors include osteogenic sarcoma and Ewing sarcoma. The orthopedic management of such tumors often involves a choice between limb salvage and amputation. Research findings suggest no difference among the quality of

life, body image, self-esteem, and social support between the two approaches in the pediatric population.^{50,75,84} However, patients with more functional lower limbs appear to have better quality of life than their peers, irrespective of whether they underwent amputation or limb salvage surgery.⁷⁵

Fibular Deficiencies, Syme, and Transtibial Amputations

Congenital fibular deficiencies are generally managed with elective amputation, orthopedic lengthening procedures, or aggressive accommodative shoe lifts. This decision is largely based on the viability of the ipsilateral foot for structural weight bearing and the extent of the limb shortening relative to the contralateral limb.³ Long-term follow-up studies in adults have failed to observe appreciable differences in health-related quality of life or physical function between the two orthopedic pathways of elective amputation and lengthening procedures.⁹⁴

In the presence of a complete or partially absent fibula, a compromised ipsilateral foot, or severe overall limb shortening, an elective Syme amputation is often performed. Because of the absence of the distal fibular condyle, the characteristic bulbous distal end of the residual limb is less pronounced. The remaining tibia often bows anteriorly and presents with a sharply defined, prominent tibial crest. As the child ages, a progressive leg length discrepancy becomes more apparent. However, this is often beneficial because it allows space for more dynamic, higher profile prosthetic foot and ankle mechanisms.

Among both transtibial and ankle disarticulation level amputations, several authors have observed a tendency in both congenital and pediatric-acquired cases for the tibia to develop angular deformities. In cases in which these deformities become pronounced enough to affect prosthetic function, they can be corrected via surgical osteotomy or hemiepiphyodesis.^{38,98}

Tibial Deficiencies and Knee Disarticulation Amputation

The management of congenital tibial deficiencies is largely dependent upon the extent of the tibial deficit and the viability of the knee extensors. In cases in which the proximal tibia is well formed and functional knee extensors are present, orthopedic management includes proximal tibial-fibular synostosis and ablation of a typically unsound residual foot with a resultant ankle disarticulation amputation in which the fibula generally serves as the long bone.^{77,82} In the absence of a proximal tibial segment and functional knee extensors, an elective knee disarticulation is generally recommended.^{77,82}

When possible, knee disarticulation amputations are preferred over transfemoral amputations in the pediatric population because this spares the distal growth plate of the femur and ultimately permits a longer, more functional residual limb through adolescence and adulthood. In addition, this amputation level retains the balance of the hip musculature, generally enables distal load bearing, and avoids the potential for postamputation terminal bony overgrowth. The greatest disadvantage associated with adult-acquired knee disarticulation amputations—the inability to match the prosthetic knee center height to the contralateral anatomic knee center—can be readily addressed in pediatric cases by correctly timing a distal femoral epiphyodesis. This shortens the residual femur creating sufficient space to use a broader range of prosthetic knee mechanisms.

Longitudinal Deficiency of the Femur, Partial

Historically referred to as PFFD, the management of partial longitudinal deficiencies of the femur (partial LDFP) can be fairly variable. The Aitken 1 classification, initially proposed in 1969, is still used to categorize the instability present in the hips of these limbs according to the presence and development of the femoral head and acetabulum.¹ Thus in addition to the inherent leg length inequality, these cases present with varying degrees of femoral malrotation, instability of the proximal joints, and inadequacy of proximal musculature. The more commonly encountered orthopedic management strategies include an elective ankle disarticulation amputation combined with knee fusion or rotationplasty. In the case of the former, the combination of the fused knee and the ablated foot creates a functional residual limb similar to that seen in a knee disarticulation amputation. Thus it is generally capable of some level of distal end bearing and distal anatomic suspension of the prosthesis. Unlike an acquired knee disarticulation, the functionality of this limb type is often affected by the variable instability at the hip joint that characterizes this limb deficiency.

In rotationplasty, the child's foot is turned 180 degrees, allowing the intact ankle joint to mimic the function of the knee. This approach has also been advocated in cases of limb salvage after tumor resection. The acceptability of this approach to individual patients has been mixed. The functionality of this procedure in response to both LDFP and tumor resection is very high,⁹² including a frequent return to sports activities in the latter group.²⁸ However, individuals have identified difficulties in psychological and emotional adjustments to the appearance of the modified limb, particularly during adolescence.³³ These negative affects must be balanced against the reality that similar challenges have also been identified with other possible orthopedic approaches.

Amputation Level and Energy Expenditure in Congenital and Acquired Pediatric Lower Limb Deficiencies

In general, in the adult population, the energy costs of walking after lower limb amputation increase as the amputation level ascends proximally up the lower limb and amputees adopt slower self-selected walking speeds.⁹⁵ However, this principle does not appear to apply to all pediatric cases.⁴⁷ As expected, among the most proximal amputation levels of transfemoral and hip disarticulation, there is a 20% to 30% reduction in self-selected walking speed coupled with an increase in metabolic energy costs approximating 150% of normal.⁴⁷ Similarly, children with bilateral lower limb amputations walk at a slightly reduced self-selected walking speed with a slightly elevated heart rate.⁴⁷ However, children with ankle disarticulations, transtibial amputations, and knee disarticulations appear to walk at essentially the same self-selected walking speeds and oxygen costs as their nonaffected peers.⁴⁷

Financial and Vocational Impacts of Pediatric Limb Loss

The cost of pediatric limb loss to individual families has only recently been addressed in the academic literature.⁹⁷ Survey data suggest that approximately 40% of these families have borne out-of-pocket expenses for prostheses with an average annual cost of \$1321. Approximately one-fourth of the surveyed families indicated that they have had an insurance claim for a new prosthesis denied. Investments of time in procuring and maintaining a

prosthesis are also high. Travel times to and from prosthetic providers were reported to be as high as 16 hours, and the time spent in the office as high as 9 hours. Cumulatively, the average time spent in transit or on site for prosthetic appointments was 5 hours per visit and 30 hours per year.⁹⁷ Additional time investments were made in the form of physical and occupational therapy, related and unrelated hospitalizations, and emergency room visits in which 25% of surveyed families reported at least one visit in the past year, and 10% reported multiple visits.⁹⁷ A separate publication observed a 60% to 150% increase in the number of hospital admissions of children with major pediatric limb deficiencies, depending on the type of deficiency.⁸⁶ Because of these time requirements, the majority of the parents of these children will make some type of work adjustment as a result of their child's limb loss, including changing their work hours, taking days off, taking extended leaves of absence, or making a job change.⁹⁷

Future Considerations

Artificial limb technology and amputation rehabilitation have experienced great advances over the past 20 years. These advances include surgical techniques that have made limb salvage and limb sparing more viable options in cases of extremity trauma. These new surgical techniques have complicated early decision making in cases of extremity trauma when both limb salvage and amputation are being considered. Additional advances include the development of artificial limb components with sophisticated microprocessor control and active power systems as well as the development of virtual reality rehabilitation training environments. These developments have improved outcomes for individuals with amputations, but they are also typically associated with increased financial costs and have created new challenges for rehabilitation providers. The appropriate application of these advances in the clinical setting requires highly specialized knowledge and skills. The prosthetic component option with the most sophisticated technology is not always the most appropriate option, and issues of insurance reimbursement and potential financial burden on the patient also need to be taken into consideration.

New developments in regenerative medicine, transplant immunology, brain-computer interfaces, and robotics are on the horizon. These advances have the potential to further revolutionize the treatment and outcomes of individuals with amputations. For clinicians to provide the highest quality care to individuals with amputations in the future, they must maintain the most up-to-date training and clinical experience with new technology and procedures. The provision of clinical services is likely to become more and more specialized over time and require provider teams with greater involvement from disciplines such as biomechanical engineering. OI represents the implantation of a porous, coated titanium implant into bones to which is attached the prosthesis. Bones that have been used include the femur, tibia, humerus, radius, and ulna. One advantage is the elimination of socket fit issues. Research in this area continues on a global basis. With these advances and future opportunities, it is anticipated that the field of amputee rehabilitation will continue to be an exciting, challenging, and rewarding area of practice.

Normal Gait

Normal human gait can appear simple and effortless to the casual observer. However, it is actually a complex phenomenon, and understanding human gait requires a solid knowledge of biomechanical

principles. Appreciation of normal gait is essential to the evaluation and management of the gait deviations seen in individuals with amputations who ambulate with the use of prosthetic limbs.

Gait Terminology

Stride: The basic unit of gait, which includes all activity between initial contact of a limb (reference limb) and the subsequent initial contact of that same limb.

Stride Length: The distance traveled during one gait cycle or stride.

Step Length: The distance traveled during one step (initial contact to end of preswing on same limb).

Step Width: The distance between the center of the feet during the double limb support portion of the gait cycle when both feet are in contact with the ground. This distance is normally 7 to 8 cm.

Cadence: The number of steps taken in a given period of time (commonly expressed as steps per minute). Average cadence during normal human ambulation is 80 to 110 steps per minute. This corresponds to an average walking speed of 60 to 80 m/min.

Stance Phase: The portion of the gait cycle during which the reference limb is in contact with the ground. During normal walking, this portion accounts for approximately 60% of the gait cycle. Stance phase includes initial contact, loading response, midstance, terminal stance, and preswing.

Swing Phase: The portion of the gait cycle during which the reference limb is NOT in contact with the ground. During normal walking, this portion accounts for approximately 40% of the gait cycle. Swing phase includes initial swing, midswing, and terminal swing.

Single Limb Support: The portion of the gait cycle during which only ONE limb is in contact with the ground. During normal gait, this segment accounts for 40% of the total gait cycle. Single limb support includes midstance and terminal stance.

Double Limb Support: The portion of the gait cycle during which TWO limbs are in contact with the ground. During normal gait, this portion accounts for 20% of the gait cycle. Double limb support includes three different components of the gait cycle: initial contact, loading response, and preswing.

Functional Tasks of Normal Gait: The functional tasks of normal human gait are typically described as weight acceptance, single limb support, and limb advancement.

Rancho Los Amigos Gait Terminology: The Rancho Los Amigos Gait Terminology is the preferred terminology to be used in the description of gait (Fig. 11.26).

Stance Phase

Initial Contact: Point in time when foot comes in contact with the ground.

Loading Response (LR): Initial contact to the time when the contralateral foot leaves the ground.

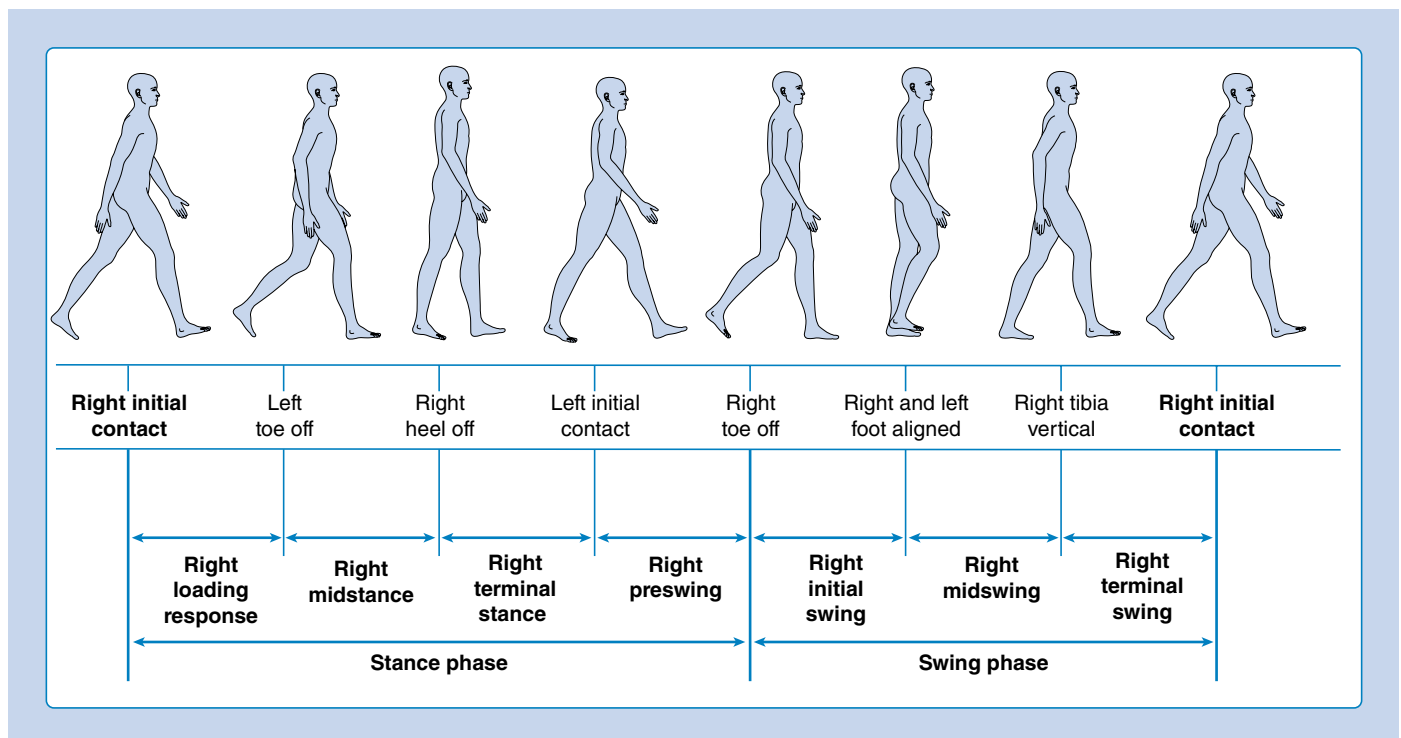
Midstance (MSt): From the time the contralateral foot leaves the ground to the time that the ipsilateral heel leaves the ground.

Terminal Stance (TSt): From the time that the ipsilateral heel leaves the ground to the time of contralateral foot initial contact with the ground.

Preswing (PSw): From the time of contralateral foot initial contact with the ground to the time that the ipsilateral foot leaves the ground.

Swing Phase

Initial Swing (ISw): The time from when the foot leaves the ground to ipsilateral foot alignment with the contralateral ankle.



• Fig. 11.26 Gait cycle and phases of gait.

Midswing (MSw): The time from ankle and foot alignment to the swing leg tibia becoming vertical.

Terminal Swing (TSw): The time from the tibia reaching a vertical position to initial contact of the swing foot with the ground.

Determinants of Gait

The six determinants of gait were originally described by Saunders et al. 1953.⁷⁶ These determinants were used to describe strategies for achieving the most efficient gait through minimizing the movement of the center of gravity (COG). As a person ambulates with a normal gait pattern, the COG follows a smooth, sinusoidal path in the frontal, transverse, and sagittal planes. The actual COG displacement is approximately 5 cm (2 inches) in each plane during normal gait. Descriptions of the determinants of gait vary in different references and have been revised over time.^{20,71,95} The determinants can be divided into three that occur at the level of the pelvis and three that occur in the knee, foot, and ankle mechanisms. The six determinants can be described as pelvic rotation in the horizontal plane, pelvic tilt in the frontal plane, lateral displacement of the pelvis, early knee flexion, foot and ankle mechanisms, and late knee flexion.

Kinematics of Gait

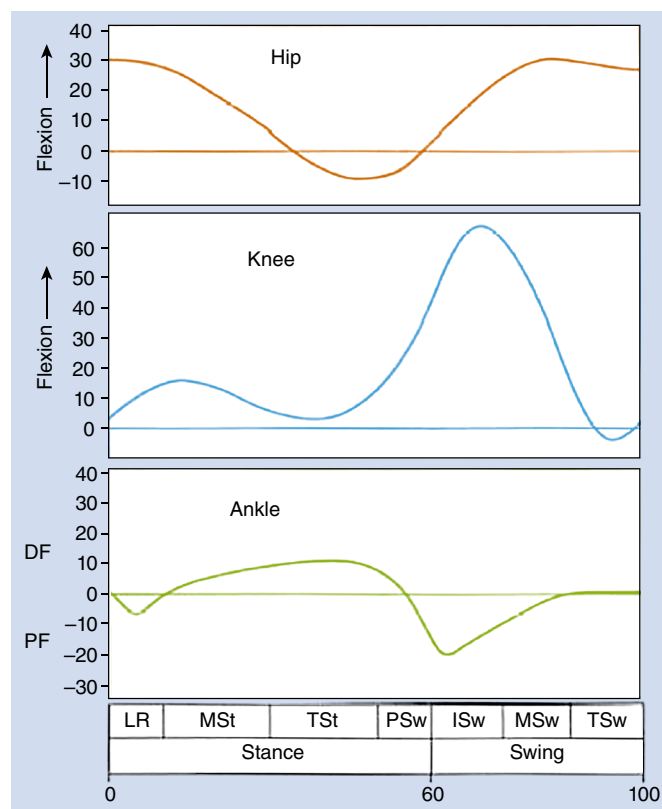
Sagittal Plane Kinematics

The sagittal plane kinematics of gait describe the motion that occurs at the hip, knee, and ankle during normal human gait in the sagittal plane^{29,64,71,95} (Fig. 11.27).

Hip: The hip begins the gait cycle in approximately 30 degrees of flexion at the time of initial contact. The hip undergoes gradual extension throughout stance phase. The hip reaches maximum extension of 10 degrees at the end of terminal stance. At the beginning of preswing, the hip begins to flex before the foot leaves contact with the ground. The hip gradually flexes during the swing phase reaching peak flexion of just over 30 degrees just before initial contact.

Knee: The knee begins the gait cycle with approximately 5 degrees of flexion at initial contact. The knee flexes slightly during loading response to 10 to 15 degrees. Once the limb is in single limb support at the beginning of midstance, the knee begins to extend, and it reaches -5 degrees of full extension before beginning a period of rapid knee flexion at the end of stance phase and into initial swing phase. During swing phase, the knee reaches maximum flexion of approximately 60 degrees during midswing before moving into a period of knee extension. The knee reaches full extension at the end of swing phase and begins to flex slightly just before initial contact.

Ankle: The ankle begins the gait cycle in a neutral position at the time of initial contact. There is rapid plantarflexion to approximately 10 degrees of plantarflexion that occurs during the loading response. This period of plantarflexion is followed by a time of gradual dorsiflexion that continues through midstance and terminal stance. Peak dorsiflexion of 10 degrees occurs just before preswing. During preswing, the ankle begins to plantarflex rapidly before the foot leaving contact with the floor. This plantarflexion continues and reaches a maximum of 20 degrees of plantarflexion at the end of stance phase and during initial swing before the ankle dorsiflexing back to a neutral position during the remainder of the swing phase.



• **Fig. 11.27** Sagittal plane kinematics for the hip, knee, and ankle during each phase of the normal gait cycle. DF, Dorsiflexion; ISw, initial swing; LR, loading response; MSt, midstance; MSw, midswing; PF, plantarflexion; PSw, preswing; TSt, terminal stance; TSw, terminal swing.

Kinetics of Gait

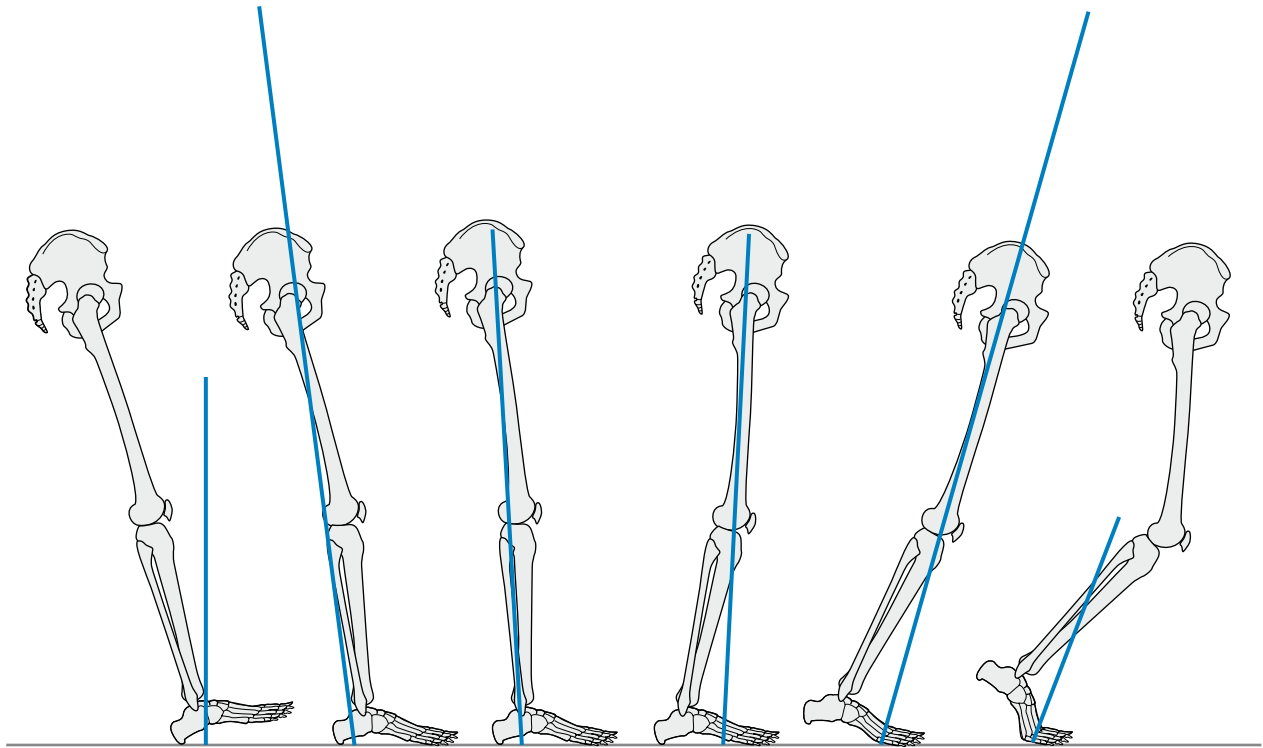
Sagittal Plane Kinetics

Sagittal plane kinetics describe the forces that occur across the hip, knee, and ankle joints during gait. Because these forces are created when the limb is in contact with the ground, they are described only during the stance phase of the gait cycle (Fig. 11.28).

Hip: The GRF is initially anterior to the hip point of rotation, which creates a flexion moment on the hip at initial contact and during the loading response. During midstance, as the tibia rotates forward, the hip moves anterior to the GRF, creating an extension moment. This extension torque across the hip remains in place throughout the remainder of stance phase, and activation of the hip flexors is required to overcome this moment in late stance to initiate hip flexion.

Knee: At initial contact, the GRF is normally located anterior to the knee, but the GRF quickly moves posterior to the knee during the loading response, which creates a flexion moment at the knee. This force must be opposed by the knee extensors to keep the knee from collapsing. The knee flexion moment remains in place until terminal stance at which time the GRF moves back anterior to the knee. At the end of preswing, just before the foot is leaving the ground, the period of rapid knee flexion begins; once the GRF moves posterior to the knee, the knee flexion moment that is created helps facilitate this motion.

Ankle: The GRF is initially located posterior to the ankle at the time of initial contact, which creates a plantarflexion moment.



• **Fig. 11.28** Location and relative magnitude of the ground reaction force vector in relation to the lower limb during stance phase of gait.

During the loading response, the plantarflexion moment continues at the ankle, and this torque has to be resisted by the ankle dorsiflexors to prevent foot drop. During midstance, the GRF moves anterior to the ankle; in terminal stance, there is a strong dorsiflexion moment at the ankle, which must be opposed by the ankle plantar flexors to limit forward progression of the tibia. In preswing, a dorsiflexion moment remains at the ankle. Thus ankle plantarflexion motion is created by concentric contraction of the ankle plantar flexors, and this helps propel the stance limb forward.

Muscle Activity During Gait

During normal gait, pelvic and lower extremity muscles generate forces to produce movement, resist GRFs, and maintain stability. The timing and extent of these muscle contractions as reported in the literature vary in different references. The information included here is meant to provide a general description of the muscle activity that is occurring in the lower extremity to either produce or inhibit motion in the sagittal plane during dynamic gait. The muscle groups listed in the following sections are either active or inactive during different phases of the gait cycle. For the purpose of this text, only the gait cycle phase in which the muscle group is active is listed (Fig. 11.29).

Hip Extensors

Late Swing: Concentric contraction to rotate the thigh posteriorly and to stabilize the limb in preparation for weight bearing in stance phase.

Early Stance: Concentric or isometric contraction to control hip and knee flexion and stabilize the limb.

Hip Flexors

Late Stance Phase: Eccentric contraction to slow and control posterior rotation (extension) of the thigh.

Swing Phase: Concentric contraction to initiate hip flexion and accelerate the swing limb forward.

Knee Extensors

Initial Contact and Loading Response: Eccentric contraction to control knee flexion and prevent knee buckling.

Late Stance and Early Swing Phase: Eccentric contraction to control collapse of the knee and prevent early heel rise.

Knee Flexors

Early and Midswing Phase: Concentric contraction in swing phase to produce knee flexion and facilitate foot clearance in swing phase.

Late Swing Phase and Early Stance Phase: Eccentric and isometric contraction to control knee extension and stabilize the limb before weight bearing.

Ankle Dorsiflexors

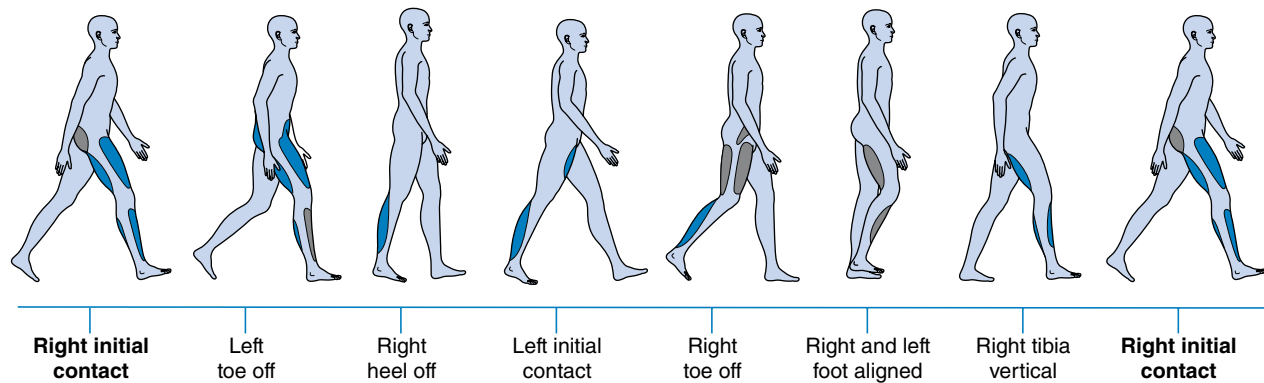
Early Stance Phase: Eccentric contraction to control ankle plantarflexion in loading response.

Swing Phase: Concentric contraction for ankle dorsiflexion and to facilitate foot clearance during swing phase.

Ankle Plantar Flexors

Midstance Phase: Eccentric contraction to control the ankle dorsiflexion moment and prevent excessive forward tibia rotation.

Terminal Stance and Preswing Phase: Concentric contraction for push-off and acceleration of the swing limb.



• **Fig. 11.29** Normal gait cycle terminology with selected limb electromyography representation. Human figures in the different phases of gait with superimposed primary muscle activity. Muscle shade intensity is roughly proportional to strength of muscle contraction.

Prosthetic Gait Deviations

The proper alignment of a lower limb prosthesis can be very challenging. There are several variables that must be considered. These include the choice of prosthetic components that have been selected for the individual patient, the position of these prosthetic knee and foot mechanisms relative to the prosthetic socket, and the extent to which each individual has learned to walk effectively with the prosthesis. In light of these considerations, it is unsurprising to note that there are a number of commonly observed gait deviations and that these may result from shortcomings in any of the considerations listed earlier, namely component selection, alignment, and patient training and walking ability. Given these complexities, many clinicians have found value in approaching prosthetic gait analysis in a systematic way. This includes analyzing gait in both the sagittal and coronal planes with consideration of each of the major joint segments across the various phases of the gait cycle.

Trans tibial Gait Deviations

Although less dramatic than those seen with transfemoral prostheses, there are a number of gait deviations that are commonly observed in patients using trans tibial prostheses.

Uneven Stride Length. Best observed in the sagittal plane, uneven stride lengths can manifest as either a shortened sound sidestep or a shortened prosthetic step. The former is more common and frequently results from a patient's lack of confidence in the prosthesis. This is especially common in newer amputees who have not yet developed confidence in their prosthesis and can improve with training and experience. By contrast, a shortened prosthetic step is frequently caused by a knee flexion contracture that prevents full extension of the knee joint in terminal swing.

Abrupt Knee Flexion In Loading Response. During loading response, a slight, smooth knee flexion is desired. There are several variables that may create a more abrupt knee flexion event. This will result if the foot is excessively dorsiflexed or if the socket is excessively flexed above the prosthetic foot. This deviation can present suddenly if a patient changes a shoe to one with a higher heel, effectively dorsiflexing the angle between the plantar surface of the shoe and the patient's knee. This will also be seen in patients who fail to eccentrically contract their quadriceps during

loading response, either because of weakness or a resultant pain created in the socket. As such, this deviation may be seen more frequently in shorter limb lengths. Finally, the relative stiffness of the heel of the prosthetic foot can also affect this deviation. Stiffer prosthetic heel mechanisms will increase the abruptness of stance phase knee flexion.

Absent Knee Flexion In Loading Response. Less frequently observed, knee flexion may also be absent in loading response. Although this may be the product of a socket that is excessively extended, a foot that is excessively plantarflexed or a prosthetic foot with excessive stiffness in the toe, it may also be a voluntary compensation by patients with weak quadriceps who are apprehensive about allowing this normal gait event to occur.

Visible pistoning. *Pistoning* is a term often used to describe a problematic movement of the residual limb in the socket. This is often best seen in the coronal plane as the limb reseats into the bottom of the socket during full weight acceptance. Depending on the type of suspension used by the patient, some degree of pistoning may be expected. For example, a visible amount of pistoning is likely to be seen with both anatomic and strap suspension systems. Pistoning may result from compromised suspension, such as a torn sealing sleeve in a suction suspension system or in a poorly fitting socket in which the volume of the socket and the limb do not match.

Coronal Knee Instability. During midstance, a modest varus moment is generally accepted at the knee. Variations are generally a product of prosthetic alignment. A prosthetic foot positioned relatively outset underneath the socket will create a valgus moment. When a prosthetic foot is positioned with an excessively inset alignment, the resultant varus moment can be excessive.

Lateral Trunk Bending. Although more commonly observed in the transfemoral population, a lateral trunk bend over the side of the prosthesis will sometimes be seen in patients using a trans-tibial prosthesis. This generally suggests a failure to fully load the prosthesis either because of socket discomfort or inadequate training. This deviation can become a fixed habit if not addressed early in prosthetic gait training.

Early/Late Heel Rise. Deviations to the timing of heel rise can occur at either extreme. An early heel rise is generally the product of excessive ankle dorsiflexion or an excessively posterior position

of the foot beneath the socket. Either misalignment effectively shortens the toe lever of the prosthesis and can result in an abrupt loading of the sound side limb. An excessively flexible prosthetic toe or keel will have the same effect. A late heel rise may be a product of an excessively plantarflexed foot, a foot that has been placed anteriorly beneath the prosthesis, or a prosthetic foot with a very rigid keel. All three scenarios will lengthen the prosthetic toe lever, making it difficult for the patient to get over the prosthetic toes during terminal stance.

Abrupt Sound Side Loading. An excessively short or flexible prosthetic toe lever will cause a patient to roll over to the toe of the prosthesis and “drop off” on to the sound side limb. This abrupt loading has been linked to elevated risks for osteoarthritis of the knee and hip. Evidence has consistently demonstrated that the dynamic resistances that characterized energy-storing and release prosthetic feet reduce the magnitude of this prosthetic drop-off and reduce the abruptness of sound side limb loading.

Sound Side Vaulting. When patients are concerned about clearing the toe of their prosthesis during swing phase, they may adapt a “vaulting” strategy, in which a concentric burst of the sound side plantar flexors raises them up on their toes, functionally lengthening the stance limb and ensuring swing phase clearance. Seen in both transtibial and transfemoral gait patterns, this deviation may suggest that the prosthesis is too long or has inadequate suspension, allowing it to functionally lengthen during swing phase. It may also be either a gait habit or preference.

Transfemoral Gait Deviations

Gait deviations at the transfemoral level are both more common and more pronounced. In contrast to the bony anatomy that characterizes most transtibial limbs, transfemoral limbs are relatively fleshy, compromising the ability of patients at this amputation level to control their prostheses through their sockets. In addition, irrespective of the various design approaches, there is always some level of instability associated with prosthetic knee mechanisms that will ultimately affect gait mechanics.

Asymmetric step lengths.

At the transfemoral level, step symmetry is often characterized by a long prosthetic step followed by a shorter sound sidestep. Particularly for the novice walker, apprehension of prosthetic knee instability creates a reluctance to fully load the prosthesis in mid-stance and terminal stance. As a result, patients tend to take shorter steps with their sound side. This habit is often formed early in rehabilitation and can persist in the absence of corrective gait training. Additionally, many patients at this level develop some level of hip flexor tightness. If this is not adequately accommodated in the alignment of the prosthesis, limitation in hip extensor mobility can also prevent patients from taking a suitable sound sidestep.

Knee instability during loading response.

The same variables that can create abrupt knee flexion during loading response in transtibial walkers can create knee instability for transfemoral walkers. In the absence of an anatomic knee joint, knee extension on the prosthetic side is a product of active hip extension and the mechanical stability properties of the knee. As a result, any factors increasing the knee flexion moment represent a threat to patient stability. Accordingly, at this amputation level, many patients prefer prosthetic feet with softer heels safely aligned in relative plantarflexion. This is especially relevant for patients with shorter limbs or weaker hip extensors.

External Foot Rotation. The fleshy nature of the transfemoral limb makes this amputation level more prone to rotational instabilities. This tendency can be exacerbated by any of the alignment characteristics identified earlier that will increase the knee flexion moment during loading response or by an improperly fitting socket. Among newer patients, this deviation may occur when the prosthesis is donned incorrectly in excessive external rotation. Less frequently, the prosthesis may simply be misaligned.

Lateral Trunk Bending. Lateral trunk bending is one of the more commonly observed deviations among transfemoral amputees. In able-bodied walkers, the hip abductors activate in mid-stance to maintain a level pelvis in single limb support. After transfemoral amputation, the ability of the prosthetic socket to stabilize the residual femur in the coronal plane is variable, generally decreasing with shorter limb lengths. In the absence of traditional coronal hip stabilization, it is very common for amputees at this level to perform a compensatory lateral trunk bend ipsilateral to the prosthesis during single limb support to bring their weight over the prosthesis. With proper training and practice, this pattern can be altered such that the lateral shift occurs at the hips rather than at the shoulders, thereby reducing the trunk compensations. However, in shorter transfemoral limbs, because of the difficulty in stabilizing the residual femur, some amount of compensatory trunk bend is to be expected.

Excessive Lumbar Lordosis. As noted, hip flexor tightness is common at this amputation level. If the alignment of the prosthesis fails to accommodate this tightness, the patient is forced to stand with excessive lumbar lordosis. Increasing the flexion angle of the socket and sliding the prosthetic knee joint posteriorly underneath the socket will accommodate hip flexor tightness and reduce the need for compensatory lordosis.

Excessive Heel Rise. The amount of heel rise seen in the transfemoral prosthesis should generally match that observed on the sound side. However, the ability to regulate this variable will depend upon the knee joint mechanism. In nonhydraulic knee joints, the friction settings of the knee can only be optimized to a single walking speed. Settings that limit heel rise at higher walking speeds will be perceived as too stiff at lower walking velocities. Conversely, friction settings that allow an appropriate amount of heel rise during self-selected speeds will allow excessive heel rise at elevated speeds. As a result, nonhydraulic knee mechanisms are generally not used for patients capable of variable speed ambulation. Hydraulic knee joints provide variable resistance to knee motion, with increasing resistance at elevated speeds. Excessive heel rise occurring with a hydraulic knee mechanism can often be addressed by altering the resistance settings of the knee.

Swing Phase Whips. The rotation of the prosthetic knee joint underneath the socket will determine the tracking of the knee joint during gait. When the prosthetic knee is set in excessive internal rotation, a lateral “whip” is observed as the heel deviates laterally with prosthetic knee flexion. Conversely, a knee joint set in excessive external rotation will create a medial “whip” as the heel of the prosthesis deviates medially with knee flexion. In many instances, whips can be reduced or eliminated by adjusting the rotational position of the knee beneath the socket. However, some patients generate rotational movements in their hips during limb advancement, creating whips that cannot be aligned out of the prosthesis.

Circumduction. Circumduction describes a pattern of hip motion in which swing phase flexion is coupled with abduction to ensure clearance of a swing leg. This compensation can occur for several reasons. Patients who are apprehensive about sagittal knee

instability may employ this strategy to reduce the need for physiologic knee flexion in swing phase. Alternately, the various extension assist mechanisms of the knee joints may be set too high. Prostheses with a very safe alignment, in which the knee joint is positioned relatively posteriorly under the socket, can make it difficult for patients to create knee flexion in swing. Alternately, the prosthesis may simply be too long. Any of these scenarios might create a situation in which circumduction is employed by the patient to ensure swing phase clearance of an inadequately shortened swing phase limb. Initially employed as a compensatory strategy, circumduction can become persistent in the absence of corrective adjustments to the prosthesis or gait training.

Vaulting. Circumduction represents one technique of facilitating swing phase clearance. Vaulting represents another. As described in the transtibial section, a rapid contraction of the sound side plantar flexors during midstance lengthens the stance limb to enable swing phase clearance. The absence of a physiologic knee joint and associated proprioceptive input is such that vaulting is observed more frequently in transfemoral gait than transtibial gait.

Excessive Terminal Impact. Terminal impact of the above-knee prosthesis can be viewed as complementary to excessive heel rise. Just as the resistance to the rising heel in initial swing can be adjusted within many prosthetic knee mechanisms, the amount of terminal impact experienced at the end of terminal swing can often be manipulated by adjusting the frictional or hydraulic settings of the knee. Many patients prefer to “feel” the terminal impact of the extending prosthetic knee at the end of each step as this increases their confidence in the sagittal stability of the prosthetic knee. Thus the amount of terminal impact preferred by individual patients can be relatively variable. Emerging evidence has suggested that the latency period between the moments when the knee reaches full extension and the heel strikes the ground is reduced among patients using the enhanced safety features associated with microprocessor-regulated knee mechanisms.

Summary

The comprehensive rehabilitation management of persons with a lower limb amputation must take into consideration issues directly related to the amputation and fitting of a prosthesis as well as the management of secondary health conditions. Optimal outcomes are achieved with the utilization of an interdisciplinary team approach and the provision of services across the continuum of care. Prosthetic restoration after lower limb amputation should be founded in understanding how to ideally match the characteristics of a prosthetic device to the functional needs of each individual patient. Advances in rehabilitation interventions and prosthetic technology have promoted individuals with lower limb amputations to return to functional independence and attain greater success in leisure activities, recreational pursuits, and competitive sports. Emerging advances in the areas of bioengineering, bionics, transplantation science, and regenerative medicine will undoubtedly lead to even greater achievements in the future.

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